Gies Sustainability Dashboard: A User-Centered Improvement

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1 Overview

The original Gies Sustainability Dashboard contains valuable SDG-aligned research data, but key insights are difficult for users to locate and interpret. To overcome this problem, our team improves and redesigns the dashboard, to make the dashboard a user-centered one, to satisfy the needs of leaders, students, and donors.

By comparing the original dashboard with our enhanced one, we demonstrate key improvements in visualizations, structure, and users' journeys. In the meantime, we have conducted a redesigned dashboard using Figma to display all the structure and charts.

The report outlines the specific details of what we have changed, why these changes matter, the structure of our new dashboard, how these changes improve users' experience, and the new users' journey based on our redesigned dashboard.

2 Method

To redesign the dashboard, we conducted a user-centered evaluation and redesign process.

2.1 Assumptions

Our redesign assumes full access to all relevant data sources, including:

- Publications from Gies and peer business schools
- All general and SDG-aligned articles
- Full faculty profiles
- Donor-impact metrics

These data sources provide the foundation for analysis and visualization.

2.2 User Group Definition

We identified three personas:

- **Leaders** — strategic, high-level decision making
- **Students** — professor matching and academic discovery
- **Donors** — evaluating impact and justifying investment

We redesign the dashboard based on three personas' experience.

2.3 User Journey Analysis & Pain Point Identifying

For each persona, we map their step-by-step flows.
At each step, we identified:

- What the user wants to do
- What action they take
- Where is the pain point of this action in the current dashboard
- What we want to modify
- How does the modification improve users' journey

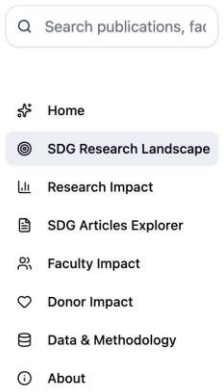
This analysis created a clear list of actionable pain points.

2.5 Generate New Users' Journey

Finally, we integrate and generate the new users' journey based on our redesigned dashboard. For each user group, we still map their new step-by-step journey. And within each step, we follow the outline:

- what they want to do
- user action
- where they click
- why it matters

3 Summary of Redesigned Dashboard



Our redesigned dashboard introduces a clearer structure Left is the new structure of our redesigned dashboard.

3.1 Parts Retained Totally or Partially

We remain the overall structure of these two sections the same, but modify some charts inside of them.

- **Research Impact** – improve the logic of how to show visualizations and modify some indexes.
    - Journal Trends: improve the logic of it, move from a broad understanding of Gies's overall research output toward a detailed view of its sustainability-oriented performance. The new logic becomes: general output → general growth → SDG share & quality → yearly structural shifts.
- **Faculty Impact** – mainly add profiles part inside of it.

3.2 Newly Added or Fully Redesigned Parts

To improve the users' experience, we add some brand new sections:

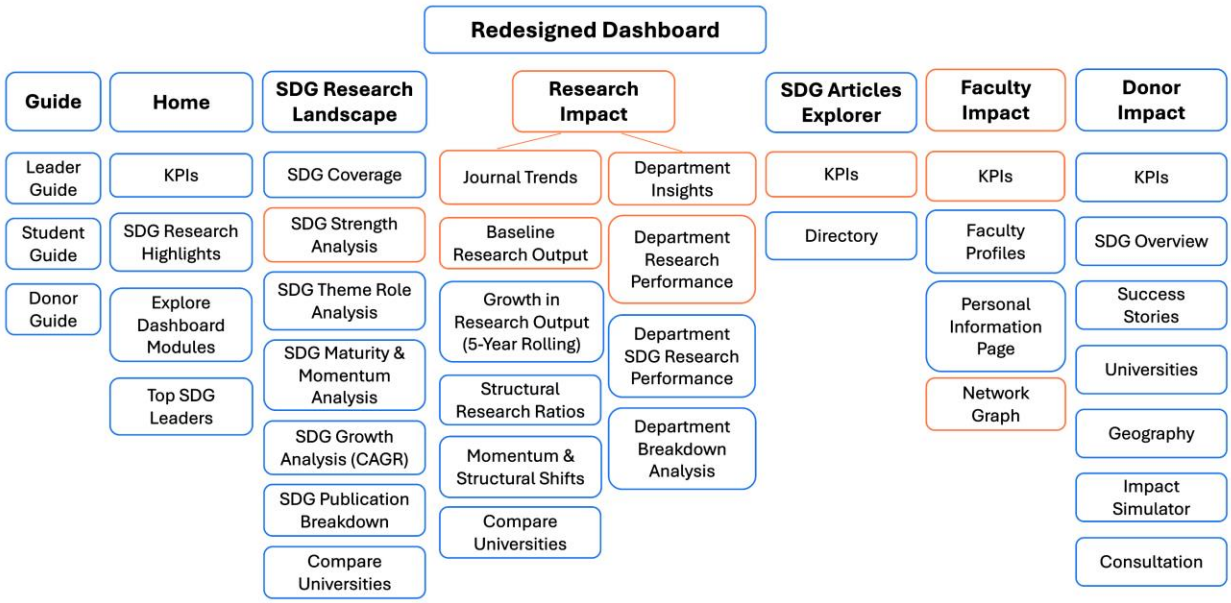
- **Guide:** provides clear step-by-step clicking sections for different users.
- **Landing Page with Key KPIs, Guidance**
  Provides a summary of Gies research situation, and some guidance blocks for users.
- **SDG Research Landscape**
  A comprehensive analytical section of SDG research of Gies, including lots of information, such as coverage, theme roles and so on.
    - Logic of this section: Coverage → Distribution → Strategic Position → Goal Momentum → Growth Rate → Breakdown.
- **SDG Articles Explorer**
  A searchable directory of all SDG articles with filters for year, SDG, department, and impact type.
- **Donor Impact Module**
  A brand new part customized for donors, with functions such as comparison across states, simulating investment impact.

- **University Comparison Function**
  Add Compare Universities parts inside two sections (SDG Landscape, Research Impact) to benchmark Gies against peer business schools across multiple SDG goals.
- **Universal Search Bar**
  Add a search bar for quick access to all the information in the new dashboard.

3.3 Redesigned Dashboard Structure

The diagram below outlines the structure of our redesigned dashboard, with seven main sections and their sub-sections. Also it shows which parts are new, which are old.

Blue parts means newly added ones, orange parts indicate ones totally or partially remained from the original dashboard.



4 Blueprint of Users' Journey

4.1 User-Specific Dashboard Module Selection

Due to different users, they require different levels of granularity. Leaders focus on high-level strategy, while students focusing on faculty and articles matching, and donors focus on return and social impact. Therefore, each persona relies on different dashboard sections.

The table below shows which sections are essential for each user group, based on their goals and decision-making needs.

Section	Specific Part	Leader	Student	Donor
Guide	Blueprint	✓	✓	✓
Home	KPIs, guidance…	✓	✓	✓
SDG Research Landscape	Coverage, Theme Role, CAGR, Strengths…	✓	✓	✓
	Compare Universities (SDG)	✓	✗	✗
Research Impact	Journal Trends	✓	✗	✗
	Compare Universities (Journal Trends)	✓	✗	✗
	Department Insights	✓	✗	✗
SDG Articles Explorer	Publications Directory	✗	✓	✗
Faculty Impact	Faculty Profiles	✓	✓	✗
	Faculty Personal Information Page	✗	✓	✗
	Network Graph	✗	✓	✗
Donor Impact	KPIs	✓	✗	✓
	SDG Overview	✓	✗	✓
	Success Stories	✓	✗	✓
	Universities	✗	✗	✓
	Geography	✗	✗	✓
	Impact Simulator	✓	✗	✓

5 Comparison between Original and New Dashboard

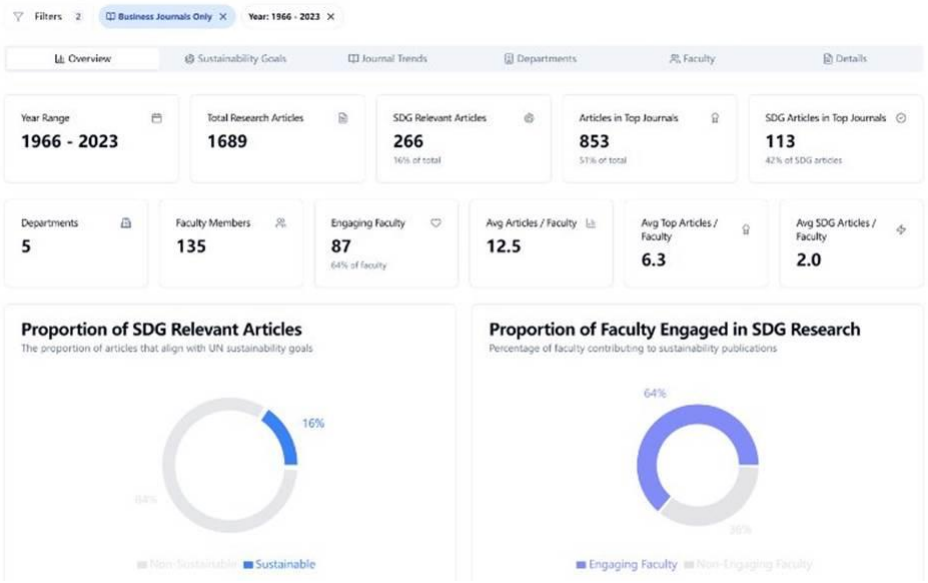
5.1 Leaders' Journey

Step 1 – Assess Gies's research situation

Pain Point 1 — The homepage lacks strategic clarity

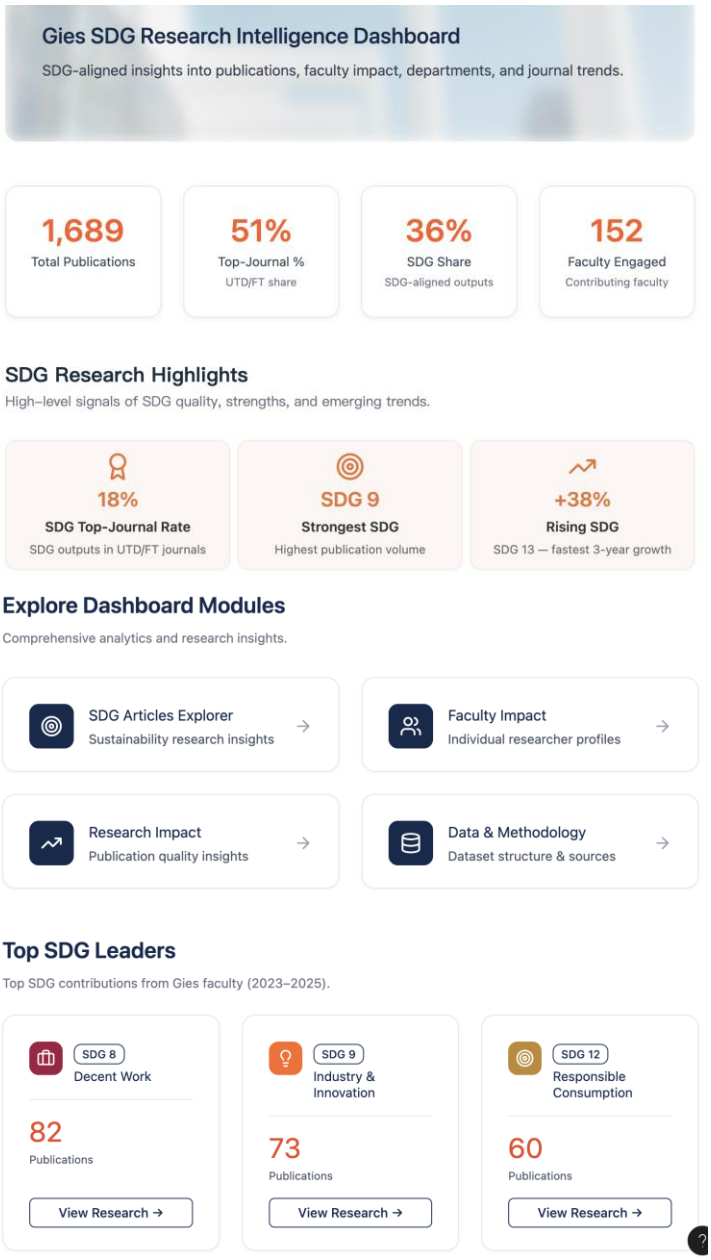
Pain Point

At the beginning of their journey, leaders want a concise overview of Gies's general and SDG research. But the original overview displayed too many information, making it difficult to identify what truly matters. Leaders struggled to quickly understand Gies's position and strategic highlights at first glance.



Solution

We redesign the Home page by introducing **4 core KPIs** and **3 SDG-aligned KPIs**, so leaders can get the overall picture at a glance. We also add navigation blocks like Explore Dashboard Modules and Top SDG Leaders to guide users quickly.



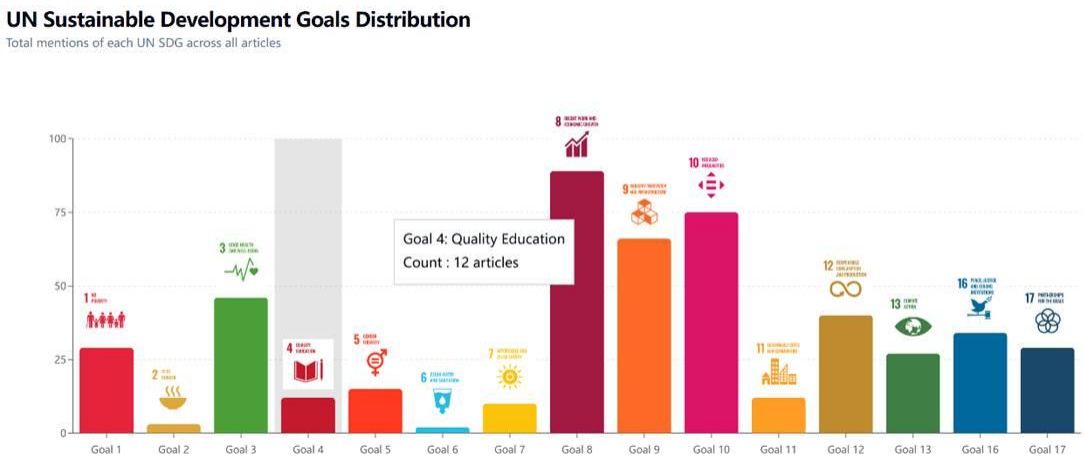
Improved Experience

Leaders can get quick and clear overview of Gies research situation, as well as the further steps' sections. The homepage becomes a **strategic briefing**, not a data overload wall.

Pain Point 2 — Leaders cannot find the positioning of each SDG

Pain Point

Leaders want to understand SDG goals performance and which goals are strength. But the previous bar charts only show basic counts without strategic insights.

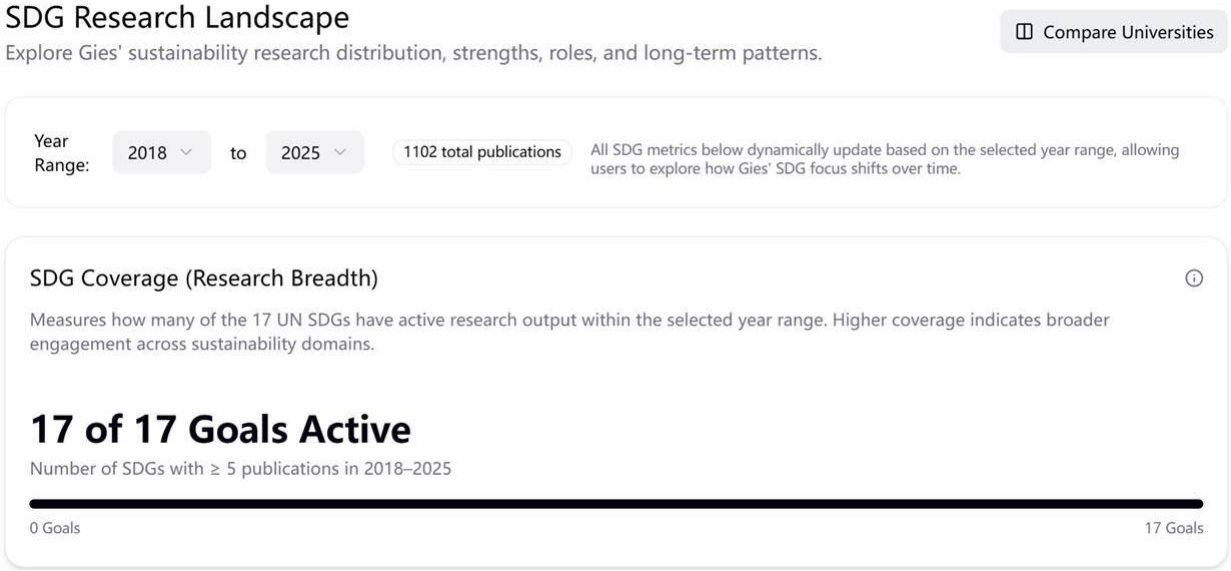


Solution

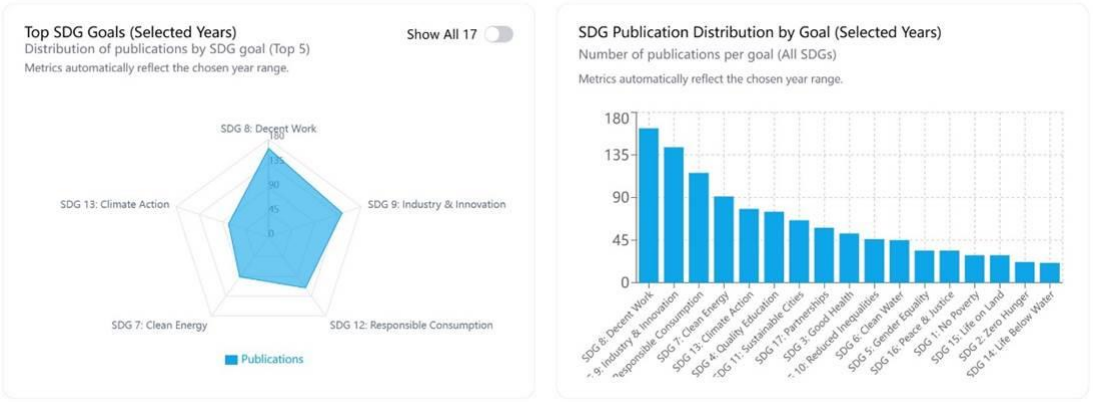
We created a new **SDG Research Landscape** page with deeper analysis:

- **Time Filter** — explore changes over time
- **SDG Coverage** — research breadth
- **SDG Strength Analytics** — top-performing goals
- **Two 4-quadrant charts** — Theme Role Analysis & Maturity & Momentum Analysis

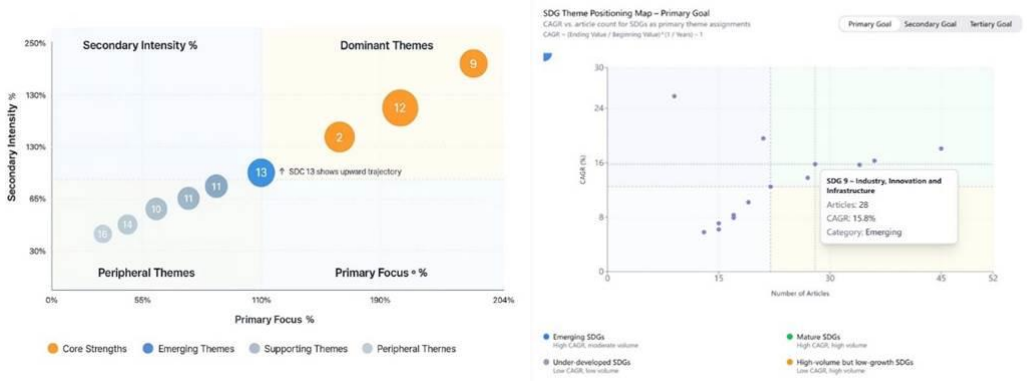
These visualizations reveal **strategic positioning** and **theme priorities**.



Time filter & SDG Coverage



SDG Strength Analytics



Two 4-quadrant charts

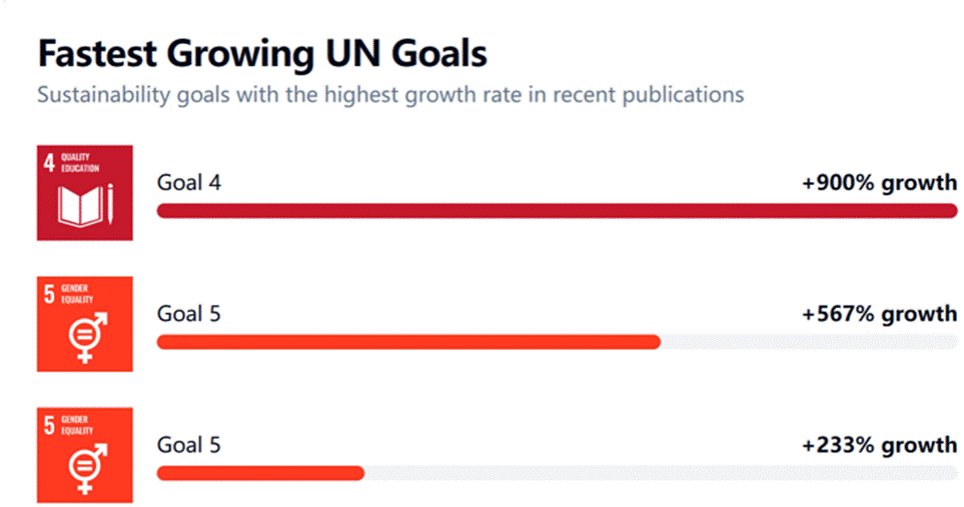
Improved Experience

Leaders can now identify: **Which SDGs lead? Which are emerging? Which lack support?**
So that leaders can have a clear insight instead of guesswork.

Pain Point 3 — Leaders cannot evaluate SDG growth and momentum

Pain Point

Leaders care about the growth of SDG goals. The original dashboard only listed “fast growing goals” which is not enough for analysis.



Solution

We added these graphs followed in the “SDG Research Impact” part:

- **CAGR heatmap** — growth visibility by color
- **CAGR Bubble chart** — combines **volume + CAGR** (it is in “Donor Impact” part, the other two charts are both in SDG Research Landscape)
- **SDG Publication Breakdown** — automatic ranking by selected year range

These enable direct identification of **high-growth + high-volume** strategic opportunities.

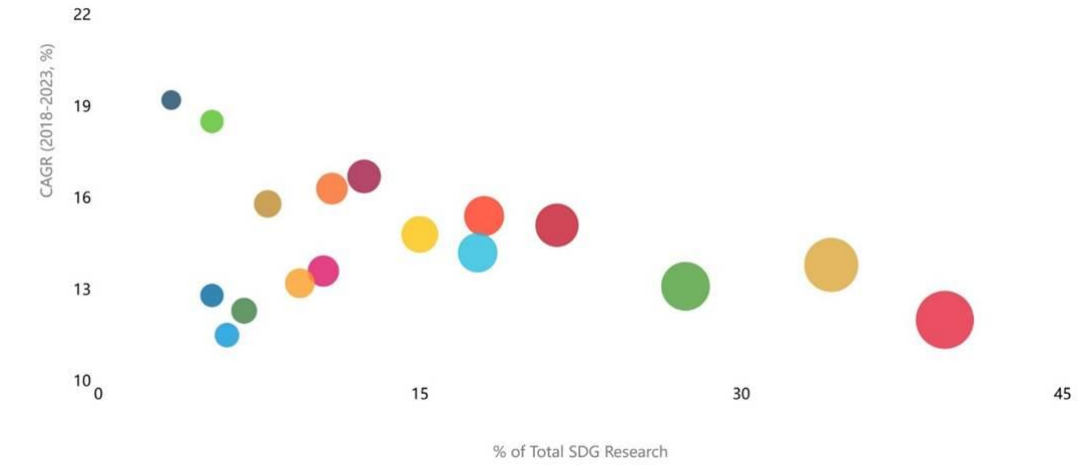


CAGR heatmap

SDG Research Landscape

Interactive visualization of all 17 SDGs. Bubble size represents publication volume, color intensity shows growth rate (CAGR).

CSV PDF



Click bubbles to filter by SDG Larger bubbles = More publications

CAGR Bubble chart in "Donor Impact"

Detailed SDG Breakdown (Selected Years)

All SDG goals for 2018–2025

Metrics automatically reflect the chosen year range.

Rank	SDG Goal	Publications (Start → End)	Percentage	CAGR %	Primary Theme Count	Secondary Theme Count	T	Tertiary Theme Count
1	SDG 8: Decent Work	12 → 31	14.8%	+14.5%	29	19		6
2	SDG 9: Industry & Innovation	10 → 28	13%	+15.8%	26	19		9
3	SDG 12: Responsible Consumption	8 → 23	10.5%	+16.3%	24	13		5
4	SDG 7: Clean Energy	6 → 18	8.3%	+17.0%	31	11		8
5	SDG 13: Climate Action	5 → 16	7.1%	+18.1%	24	16		4
6	SDG 4: Quality Education	5 → 15	6.8%	+17.0%	16	16		11
7	SDG 11: Sustainable Cities	4 → 14	6%	+19.6%	12	23		13
8	SDG 17: Partnerships	3 → 13	5.3%	+23.3%	17	23		8
9	SDG 3: Good Health	3 → 10	4.7%	+18.8%	10	18		13
10	SDG 10: Reduced Inequalities	3 → 9	4.2%	+17.0%	14	25		5
11	SDG 6: Clean Water	3 → 9	4.1%	+17.0%	13	15		10
12	SDG 5: Gender Equality	2 → 7	3.1%	+19.6%	17	26		6
13	SDG 16: Peace & Justice	2 → 7	3.1%	+19.6%	7	9		16
14	SDG 1: No Poverty	2 → 6	2.6%	+17.0%	3	17		19
15	SDG 15: Life on Land	2 → 6	2.6%	+17.0%	6	8		15
16	SDG 2: Zero Hunger	1 → 4	2%	+21.9%	9	10		17
17	SDG 14: Life Below Water	1 → 5	1.9%	+25.8%	5	17		29

Rank automatically recalculates based on the selected year range and sorting preference.

SDG Publication Breakdown-Detailed SDG Breakdown

SDG Publication Breakdown
Comprehensive metrics for all 17 Sustainable Development Goals

Export CSV

SDG	Goal Name	Publications	% of Total	CAGR	Policy Impact	Trend
8	Decent Work	105	39.5%	12%	45	Stable
9	Industry & Innovation	91	34.2%	13.8%	52	Moderate
12	Responsible Consumption	73	27.4%	13.1%	38	Moderate
7	Clean Energy	57	21.4%	15.1%	41	High Growth
13	Climate Action	48	18%	15.4%	41	High Growth
4	Quality Education	47	17.7%	14.2%	28	Moderate
11	Sustainable Cities	40	15%	14.8%	35	Moderate
3	Good Health	33	12.4%	16.7%	38	High Growth
10	Reduced Inequalities	29	10.9%	16.3%	31	High Growth
6	Clean Water	28	10.5%	13.6%	24	Moderate
17	Partnerships	25	9.4%	13.2%	27	Moderate
5	Gender Equality	21	7.9%	15.8%	19	High Growth
1	No Poverty	18	6.8%	12.3%	15	Stable
16	Peace & Justice	16	6%	11.5%	21	Stable
2	Zero Hunger	14	5.3%	18.5%	22	High Growth
15	Life on Land	14	5.3%	12.8%	16	Stable
14	Life Below Water	9	3.4%	19.2%	12	High Growth

SDG Publication Breakdown-SDG Publication breakdown

Improved Experience

Leaders can see trend in a more visual and actionable way**.**

Pain Point 4 — Unable to understand overall SDG research trends of Gies

Pain Point

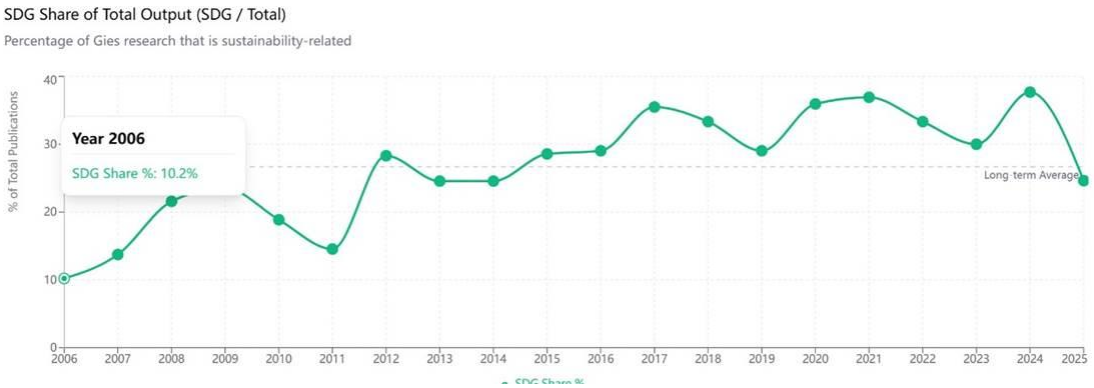
The current dashboard does not have enough trend data, so leaders can not deeply analyze whether sustainability research is improving or declining in either quantity or quality.

Solution

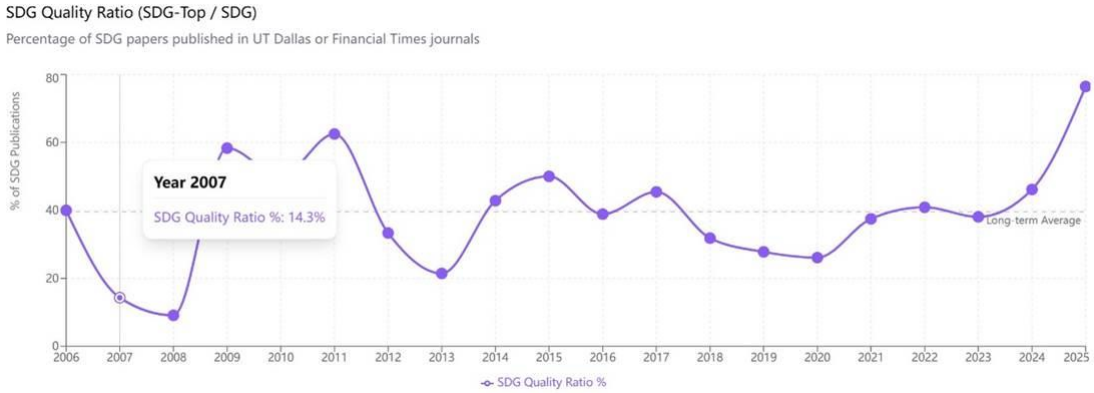
We add **Structural Research Ratios** and **Momentum & Structural Shifts** in Research Impact-Journal Trends:

- SDG Share of Total Output
- SDG Quality Ratio
- Top-tier Concentration
- YoY change metrics to reveal structural evolution

These charts show how research **intensity** and **quality composition** shift over time.



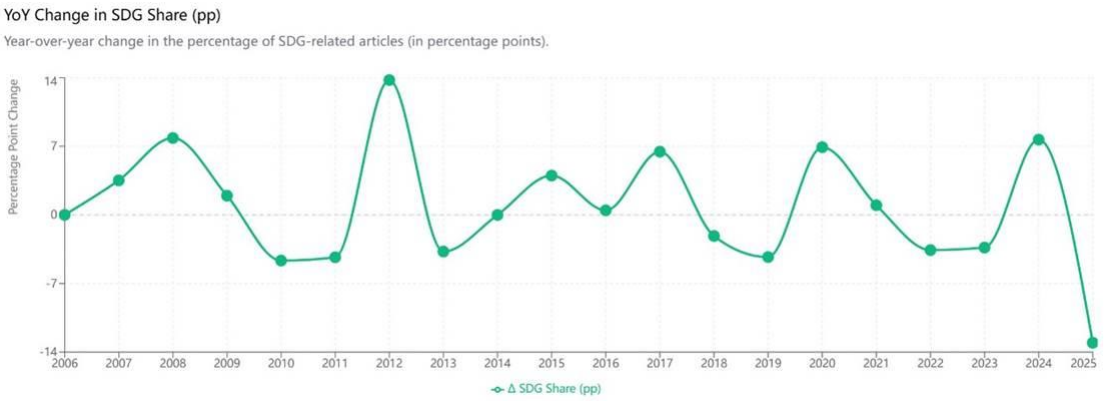
SDG share of total output (SDG/Total)



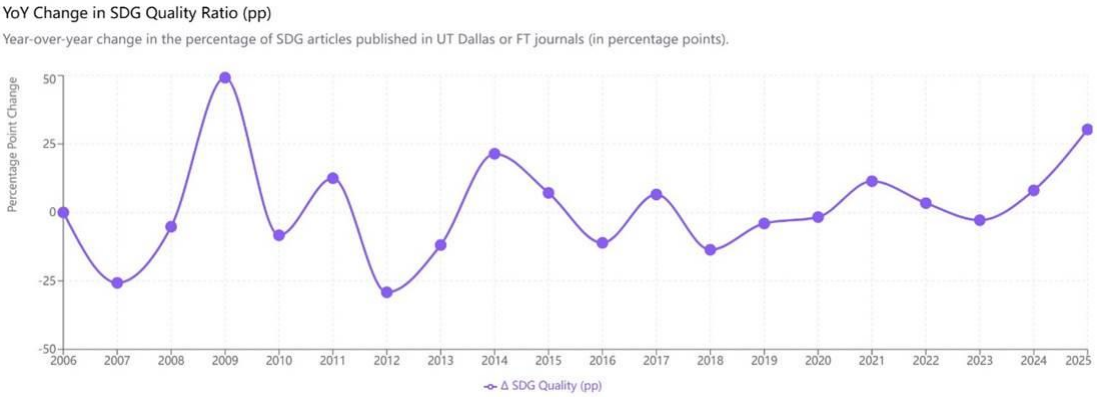
SDG quality ratio (SDG_Top / SDG)



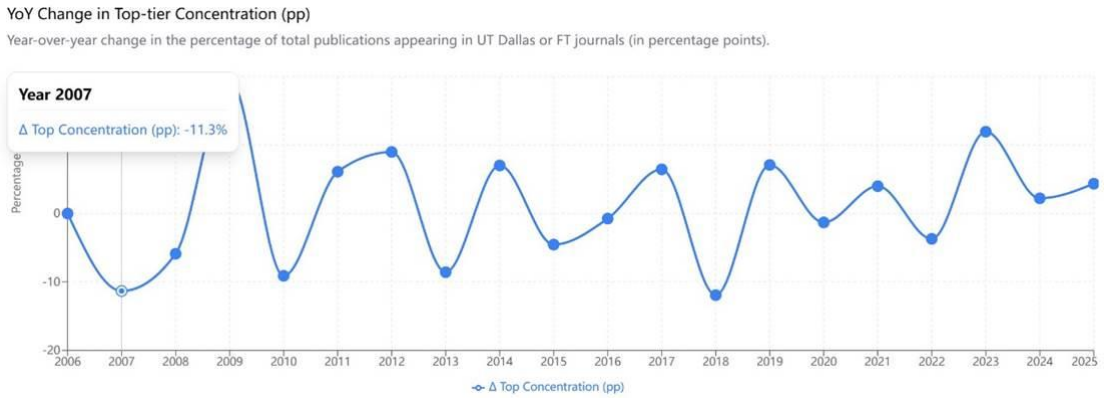
Top tier concentration (Top / Total)



YoY Change in SDG Share (pp)
*(*pp means "current year minus previous year")*



YoY Change in SDG Quality Ratio (pp)



YoY Change in Top-tier Concentration (pp)

Improved Experience

Leaders can now judge trajectory and clarify whether current strategies are working.

Step 2 – Strategic Benchmarking

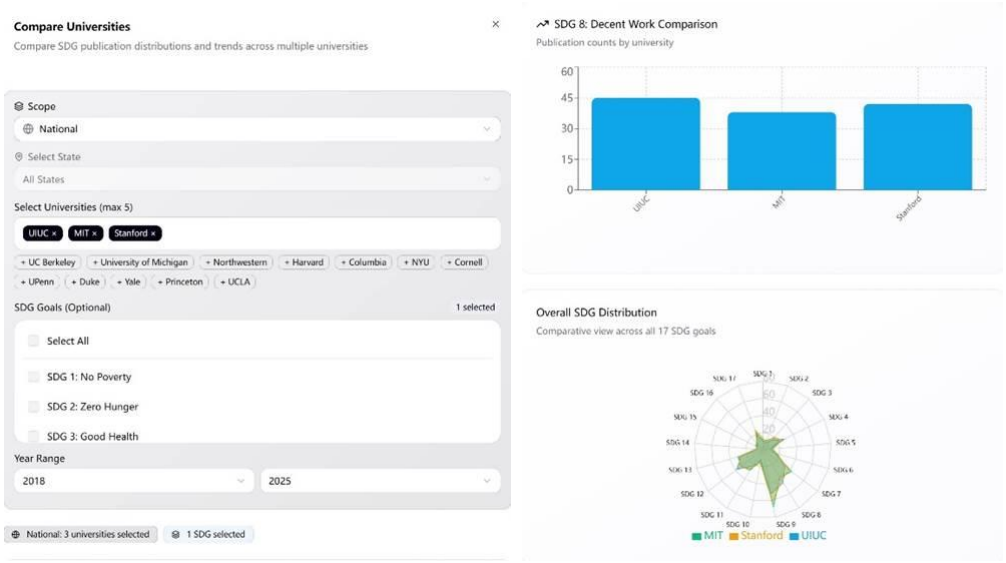
Pain Point 1 — No external benchmarking for sustainability positioning

Pain Point

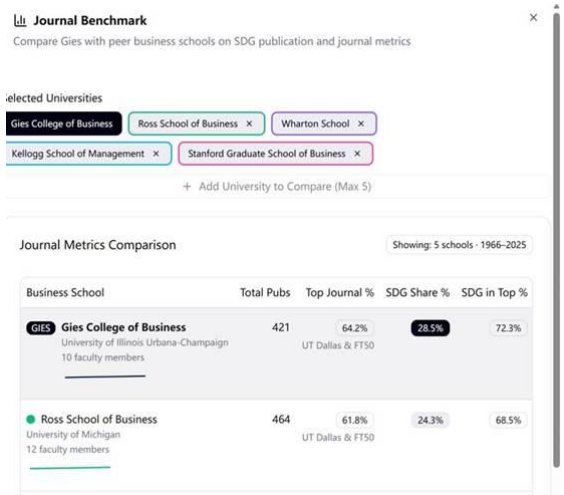
After understanding internal SDG performance, leaders need to know **where Gies stands in the competitive landscape**. The original dashboard only shows internal indicators without any peer comparison. Leaders could not evaluate whether Gies was leading, average, or lagging in sustainability research.

Solution

We introduce **Compare Universities (University Filter),** benchmarking with peer institutions on total sustainability publications, output structure, and SDG-aligned performance. Leaders can choose schools to compare and see where Gies leads or falls behind.



Compare universities- SDG Research Landscape



Compare universities- Research Impact- Journal Trends

Improved Experience

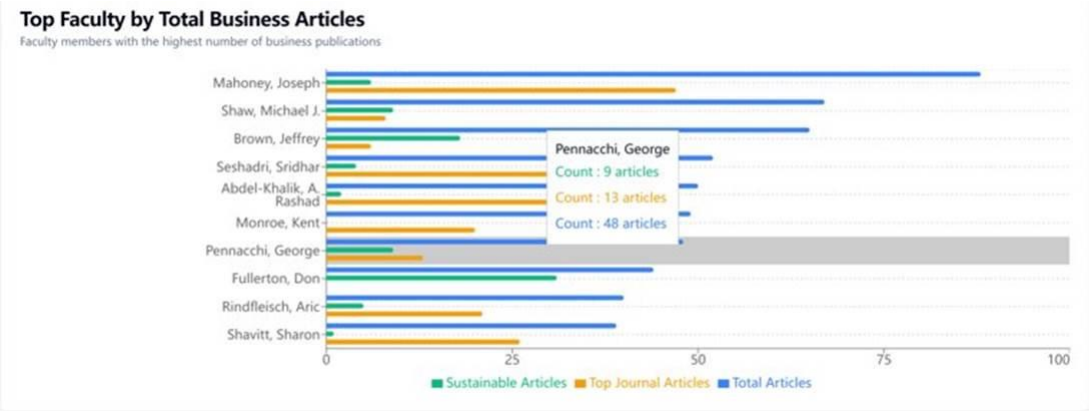
Leaders can compare Gies with other business schools to see Gies's position in the US, and also gain a contextualized view instead of isolated numbers, easy to see the clearer recognition of gaps.

Step 3 – Performance Identification

Pain Point 1 — Hard to identify which faculty contribute most to sustainability

Pain Point

Once leaders know Gies's position among peers, they want to see which faculty truly drive sustainability impact. However, the old dashboard evaluates faculty mainly by total publications and some other volumes, with little insights and can not see the impacts of each faculty directly.

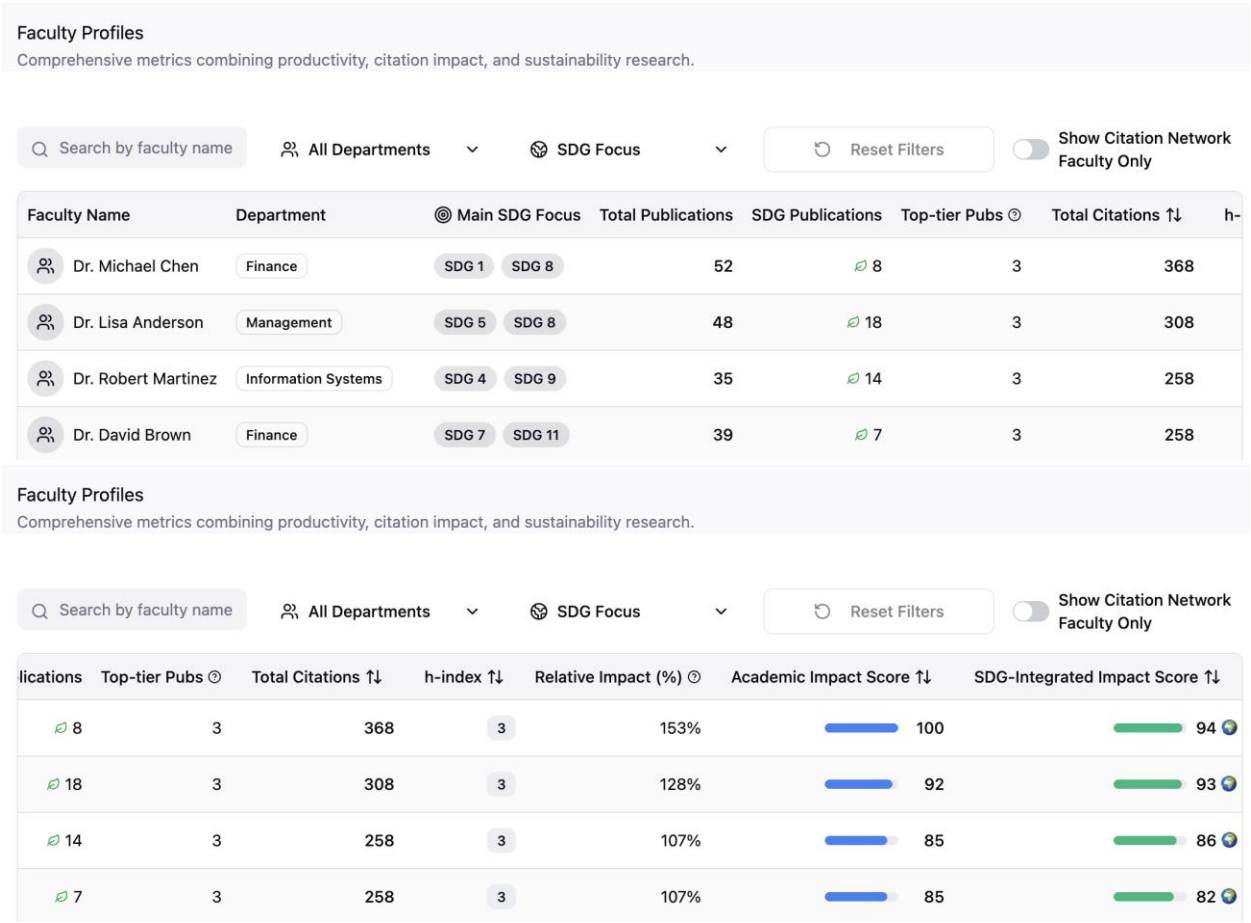


Solution

We add a new part named **Faculty Profiles** and several metrics, such as h-index, relative-impact, academic and SDG-integrated impact score, which emphasize:

- **SDG-related publication counts**
- **Thematic contribution to key sustainability areas**
- **Sustainability engagement**

This highlights faculty who meaningfully support sustainability impact.



list of faculties according to the impact score and SDG impact score

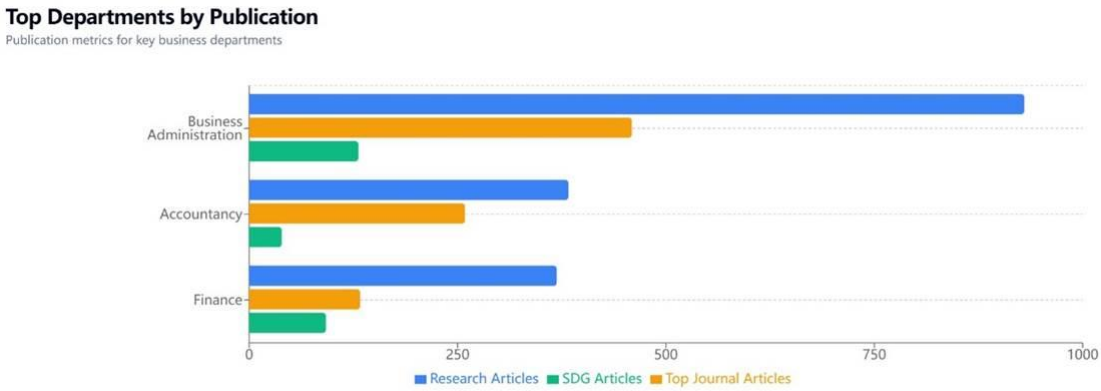
Improved Experience

Leaders gain a fairer and more impact-oriented view of talent strengths, enabling targeted support for the right faculty.

Pain Point 2 — Lack of insight into departmental direction and strategic roles

Pain Point

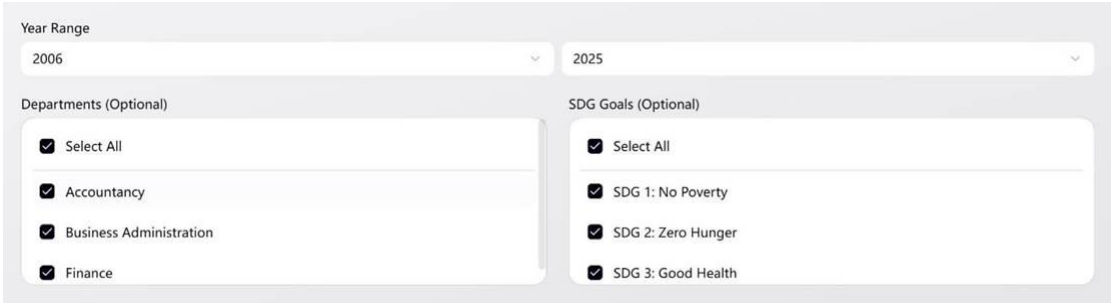
Resource allocation also depends on identifying **which departments** hold strategic importance in sustainability research. But the previous dashboard only shows publication totals with little SDG perspective, which is hard to identify differences, specialization, and growth potential across departments.



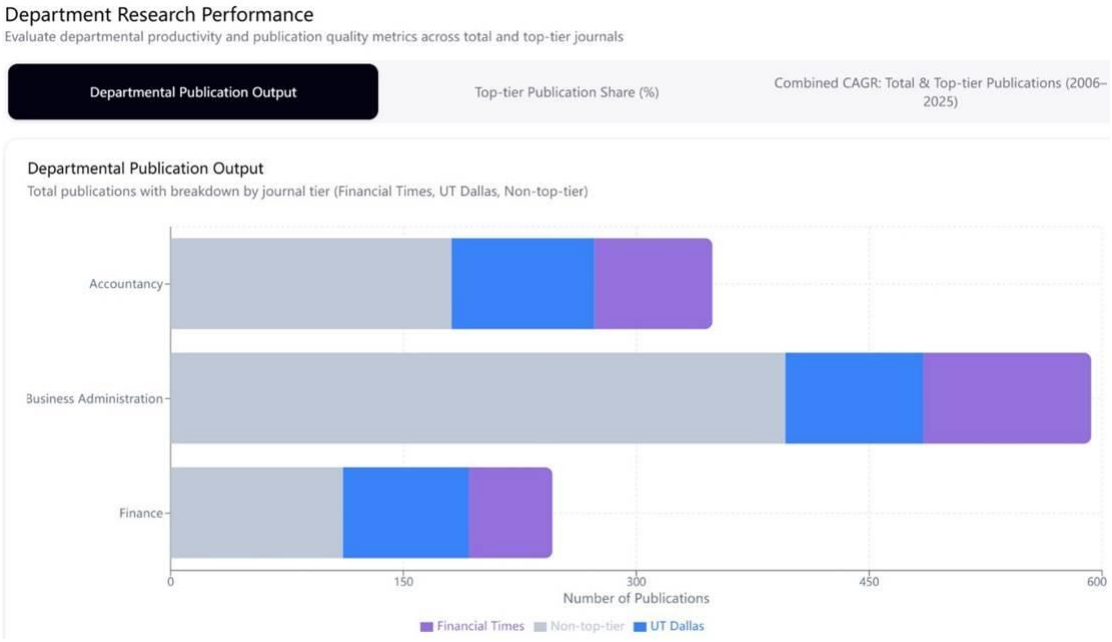
Solution

We add more **SDG-aligned metrics** in Department insights to reveal:

- Which departments already lead sustainability effort
- Which have emerging potential
- Which require support or strategy adjustment



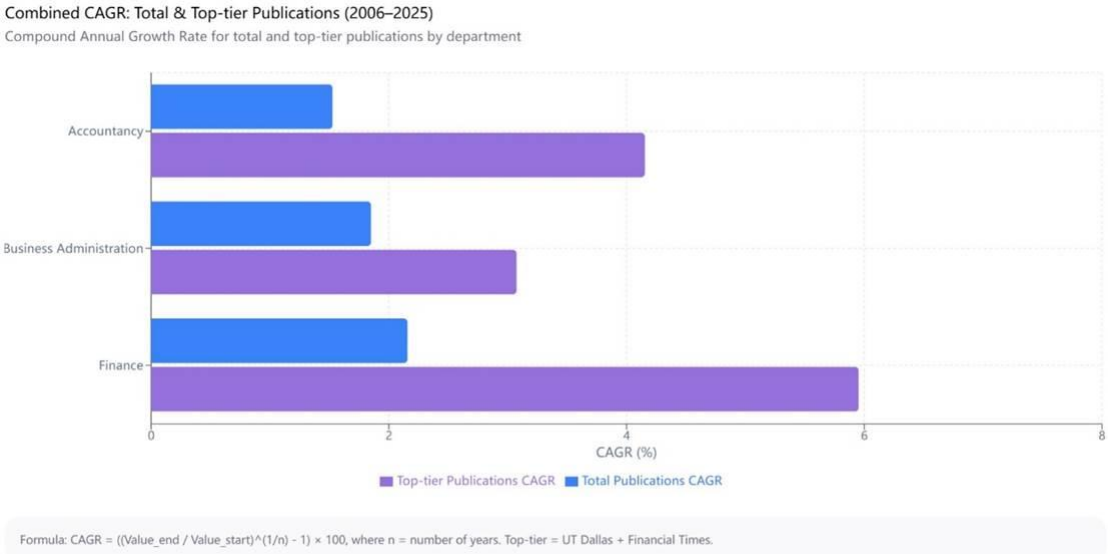
Year, Department, SDG filter on the top of the "department insights" part



Department Publication Output



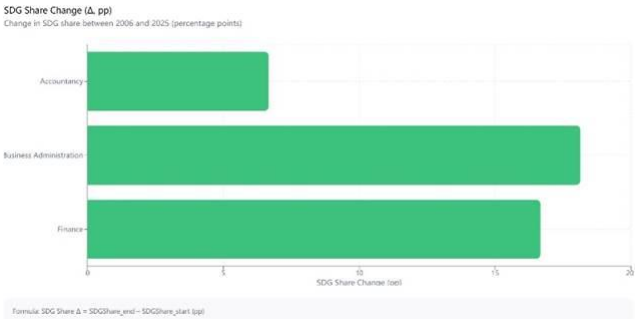
Top-tier Publication Share



Combined CAGR: Total & Top-tier Publications



SDG Publications by Department & SDG Publication Share (%)



SDG Share Change (Δ, pp)

SDG Focus Heatmap

Visual representation of SDG specialization intensity by department with publication counts

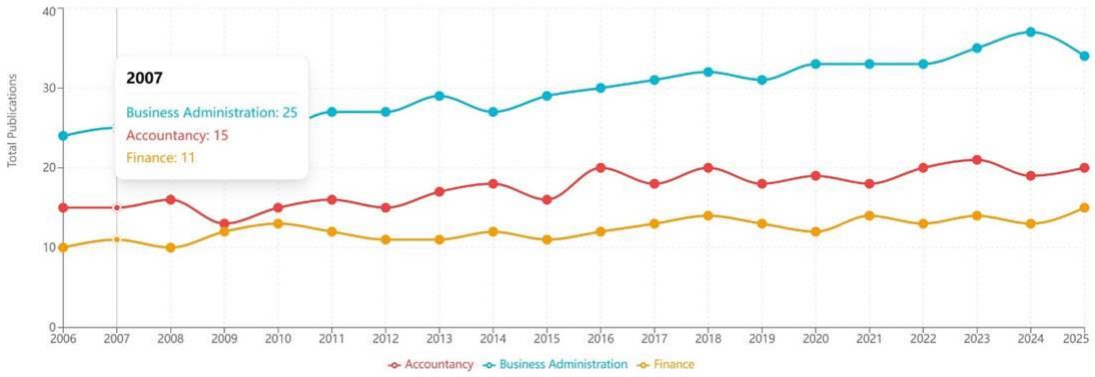
Department	SDG 3	SDG 7	SDG 8	SDG 9	SDG 10	SDG 11	SDG 12	SDG 13	SDG 16
Business Administration	7.7% (15 pubs)	13.4% (26 pubs)	11.9% (23 pubs)	2.6% (5 pubs)	10.3% (20 pubs)	14.9% (29 pubs)	12.9% (25 pubs)	11.3% (22 pubs)	2.1% (4 pubs)
Accountancy	4.4% (4 pubs)	4.4% (4 pubs)	18.9% (17 pubs)	4.4% (4 pubs)	3.3% (3 pubs)	7.8% (7 pubs)	37.8% (34 pubs)	4.4% (4 pubs)	14.4% (13 pubs)
Finance	4.9% (2 pubs)	4.9% (2 pubs)	9.8% (4 pubs)	24.4% (10 pubs)	9.8% (4 pubs)	9.8% (4 pubs)	2.4% (1 pubs)	31.7% (13 pubs)	2.4% (1 pubs)

Color intensity represents the percentage of that department's SDG publications focused on each goal. Publication counts are shown in parentheses. Colors match official SDG palette.

SDG Focus Breakdown - SDG Focus Heatmap

Annual Trend: Total Publications (2006-2025)

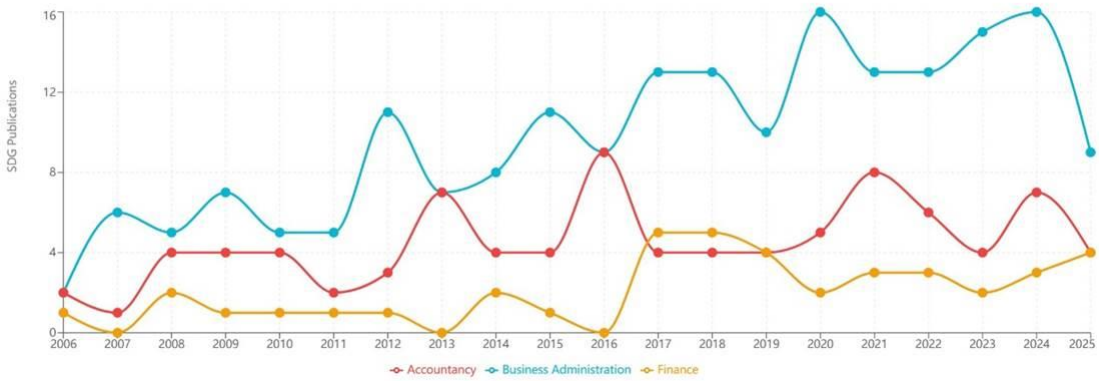
Year-over-year publication trends by department



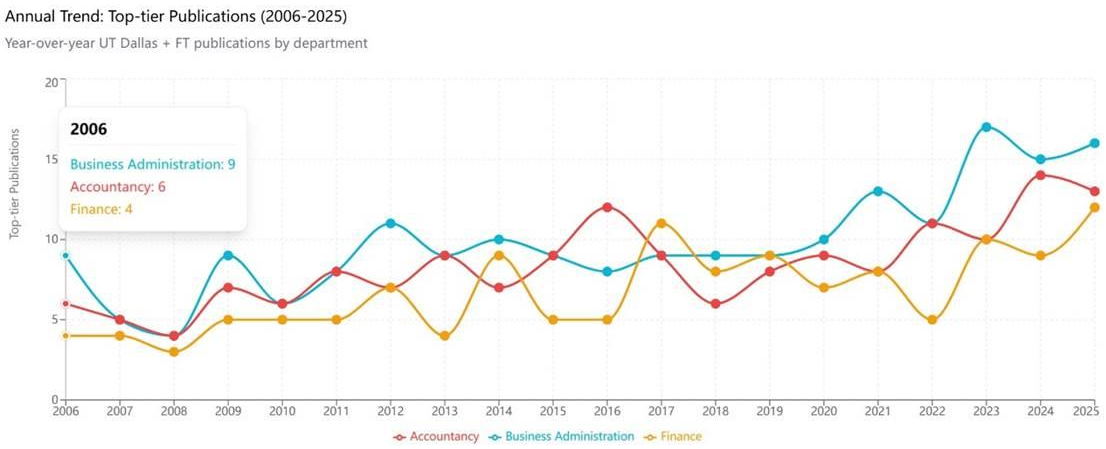
Annual Trend: Total Publication

Annual Trend: SDG Publications (2006-2025)

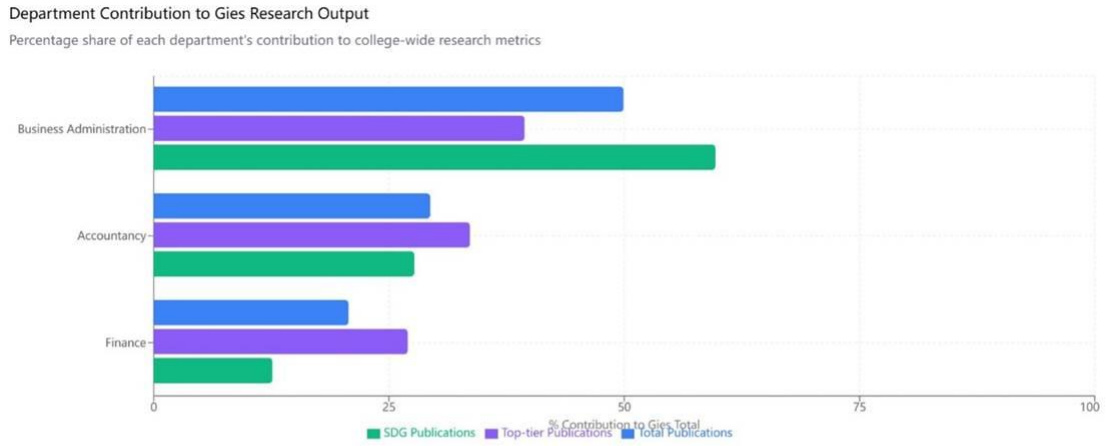
Year-over-year SDG-aligned publications by department



Annual Trend: SDG Publications



Annual Trend: Top-tier Publications



Summary Statistics

Department	Total Pubs	% of Gies Total	Top-tier Pubs	% of Gies Top	SDG Pubs	% of Gies SDG
Business Administration	593	49.9%	197	39.4%	194	59.7%
Accountancy	349	29.4%	168	33.6%	90	27.7%
Finance	246	20.7%	135	27.0%	41	12.6%
Gies Total	**1188**	**100.0%**	**500**	**100.0%**	**325**	**100.0%**

Department Contribution to Gies Total

Improved Experience

Leaders can now ensure the resources go to the departments whose strategic value and readiness are highest.

Step 4 – Narrative Validation

Pain Point 1 — No success cases to support external narrative & funding justification

Pain Point

In this step, leaders want to see **real, occurred success stories** to show policy adoption, real-world outcomes, and funding attraction of Gies. These stories are critical evidence for

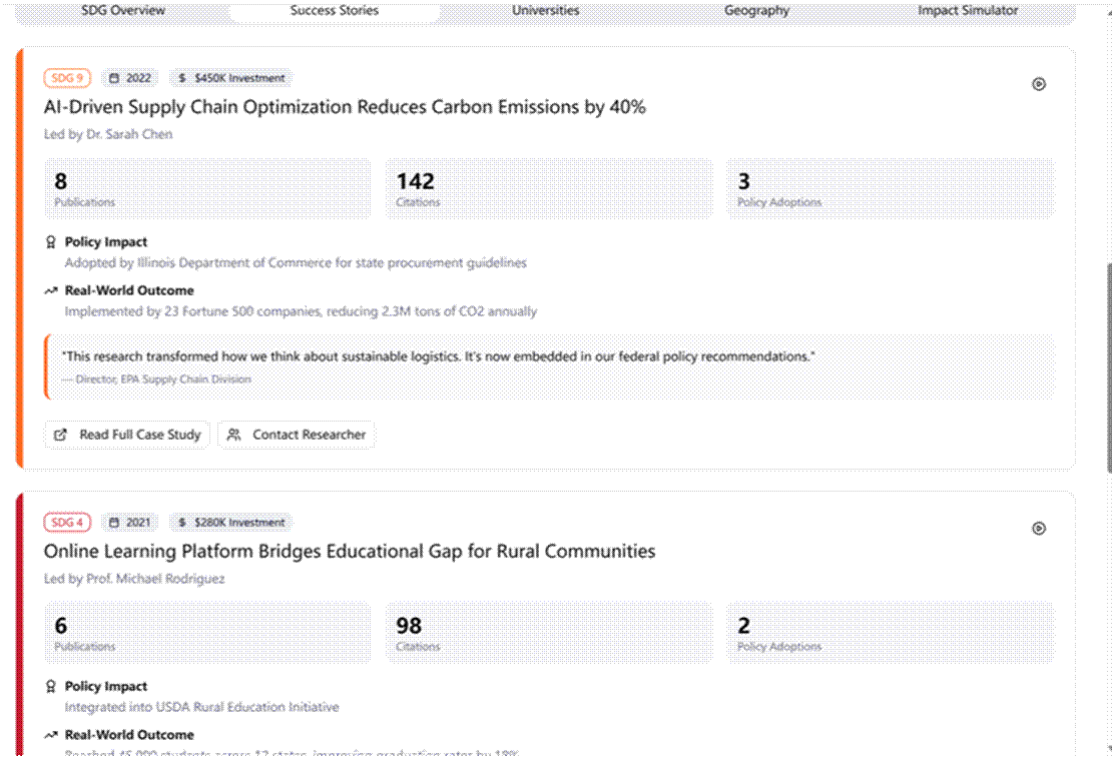
external reputation, fundraising, and impact communication. However, the original dashboard does not provide any **visible success stories.** Leaders were unable to translate analytics into convincing storytelling for donors or partners.

Solution

We add **Success Stories** in the Donor Impact section to reveal:

- Policy impact indicators
- Real-world outcome descriptions
- Funding amount & year of recognition
- Contact researcher & detailed case links

Leaders can now easily extract high-confidence narratives for communication and fundraising.



Success story part in "donor impact"

Improved Experience

There're real evidence for the strategic decision. Leaders can show the proof to display what Gies has done, which turns insights into influence.

Step 5 – Simulated Outcome Assessment

Pain Point 1 — No way to assess investment outcomes or justify resource allocation

Pain Point

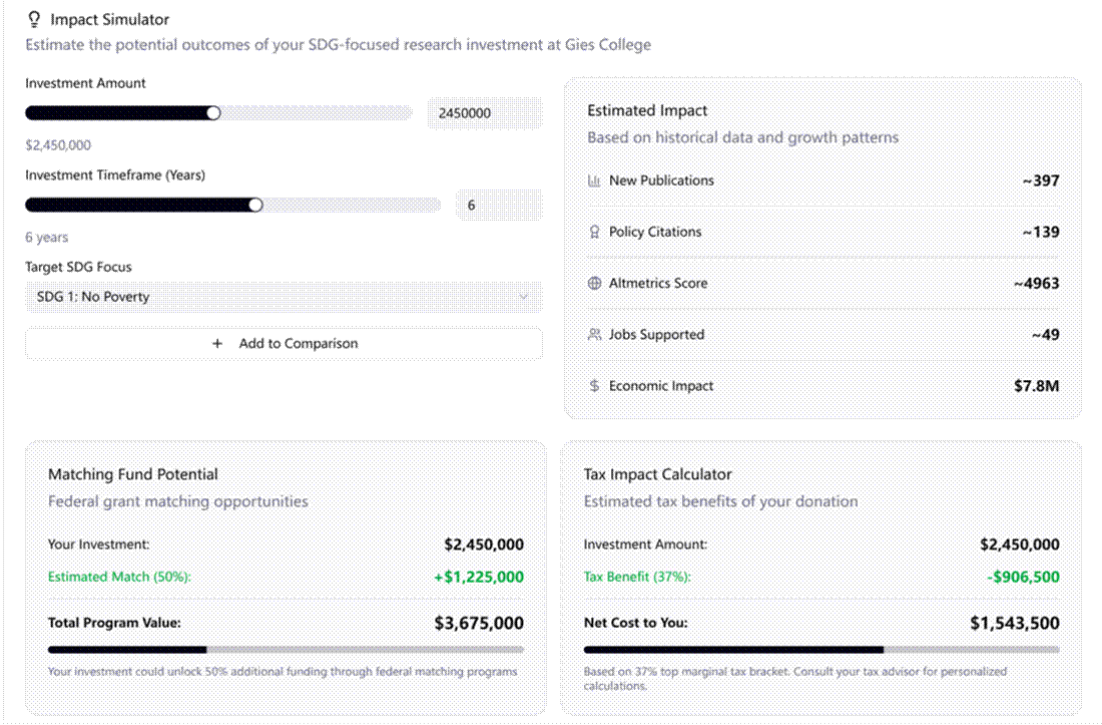
Before allocating funding, leaders must estimate potential **returns of investment** to confirm how many resources should be assigned to each project or SDG area. However, the original dashboard provides **no simulation capability.** Leaders could not forecast expected publications, policy impact, or economic outcomes based on different budget levels or time range. Decisions were based on assumptions rather than evidence.

Solution

We add the **Impact Simulator** in the Donor Impact page to support:

- Budget adjustment function — leaders can change investment amount & time range
- Estimated impact function — auto-generated projections for:
  - Publications
  - Policy citations
  - Altmetrics score
  - Jobs supported
  - Economic impact ($)

These estimates are driven by **historical patterns and growth analytics**.



"budget adjustment function" & "estimated impact function"

Improved Experience

Now leaders can see how much impact their investment could unlock before spending a dollar. This part enables evidence-based and stronger justification for funding requests.

Step 6 – Resource Prioritization & Final Verification

Pain Point 1 — No robust way to finalize resource prioritization

Pain Point

Once potential impact has been evaluated, leaders must **make the final allocation decision**: which SDG areas and researchers should receive funding first. However, the original dashboard provides **limited decision-support data**, which makes it difficult to verify whether the chosen priorities were justified. Leaders lack a **cross-check mechanism** to validate investment choices confidently.

Solution

Leaders now can do confident, cross-verify decisions through multi-angle outputs from the previous five steps, including SDG strengths, competitive status, key contributors, narrative validation evidence, and simulated outcomes.

5.2 Students' Journey

Step 1 – Assess Gies's research situation

Pain Point 1 — The homepage lacks strategic clarity

Pain Point

Students want to quickly understand Gies's overall research profile, but the original dashboard overwhelms them with dense KPIs, fragmented cards, and scattered visualizations.

Although information is technically present, the lack of hierarchy makes it hard for students to extract the "big picture" — **Which SDGs matter most? Where are the strengths? What should they pay attention to first?**

As a result, students struggle to build an initial mental model of Gies's sustainability research landscape.

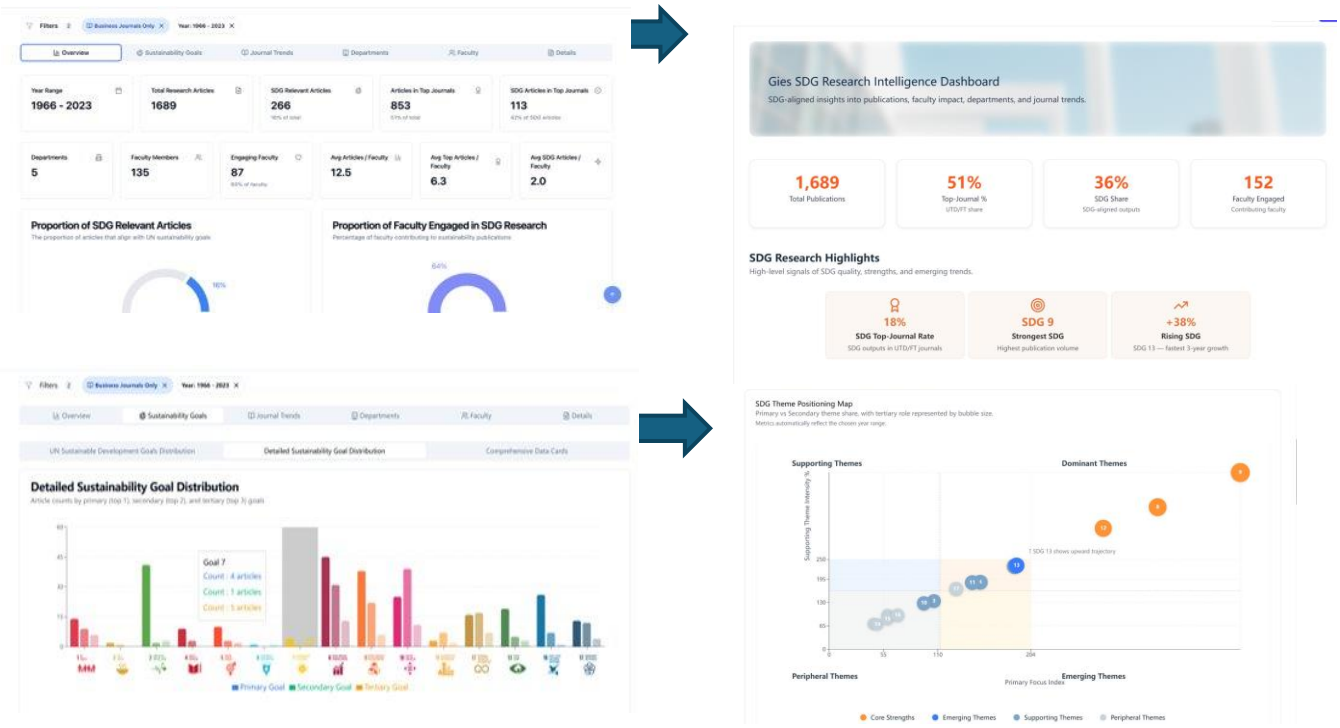
Solution

Similar to leader's journey step1, the Home and SDG Research Landscape have been created.

The key performance indicators have been integrated, and the advantages of the sustainable development goals have been presented in an intuitive form. Meanwhile, the complex performance indicator sections have been replaced by concise summaries.

This enables students to gain:

- total research scale
- strongest SDG themes
- fastest growing SDGs
- how Gies positions itself across the SDG landscape



Improved Experience

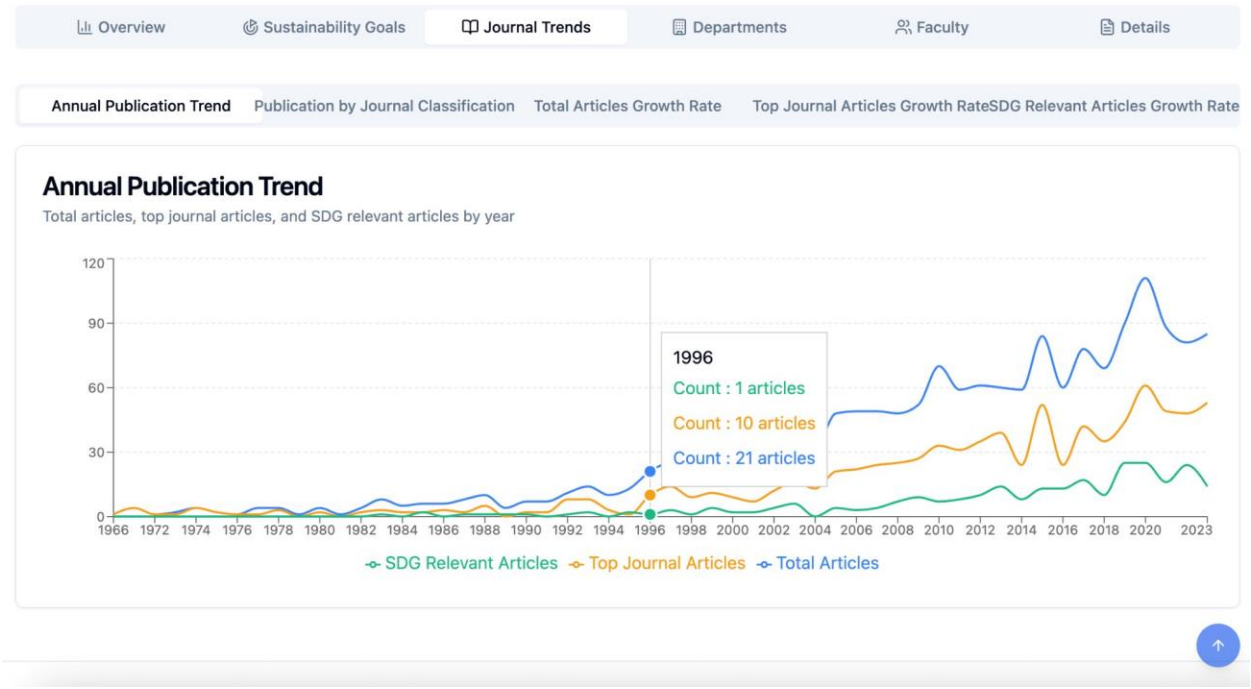
Students can now quickly grasp Gies's general and SDG research landscape through a simplified, insight-driven overview, as well as a integrated page for SDG research.

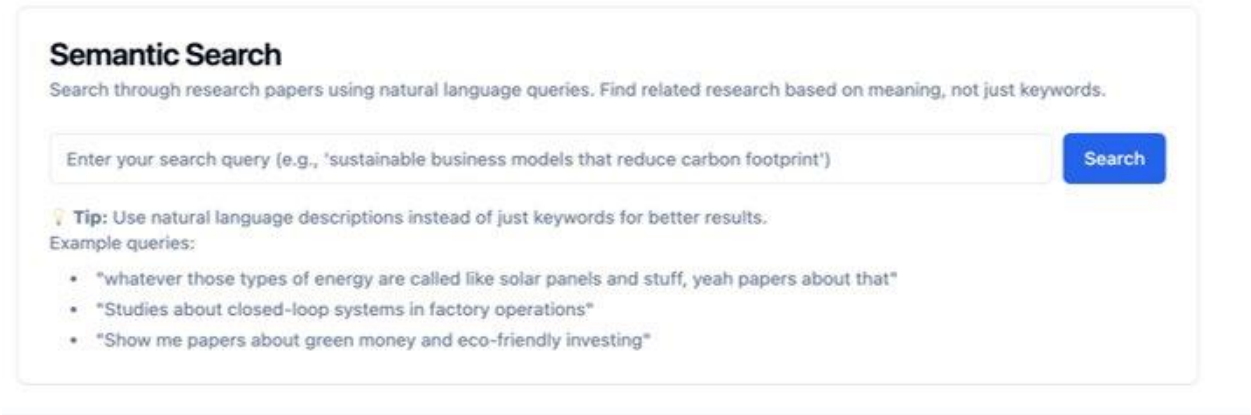
Step 2 – Article Exploration

Pain Point 1 — Students cannot access or locate actual research articles

Pain Point

Students want to read real publications tied to specific SDGs or topics. But the current dashboard only shows the trend with some line charts, but without a total SDG articles list as a directory with authors, titles, SDG tags, metadata, and way to open any article.
They cannot find, filter, or read the actual research behind the numbers.



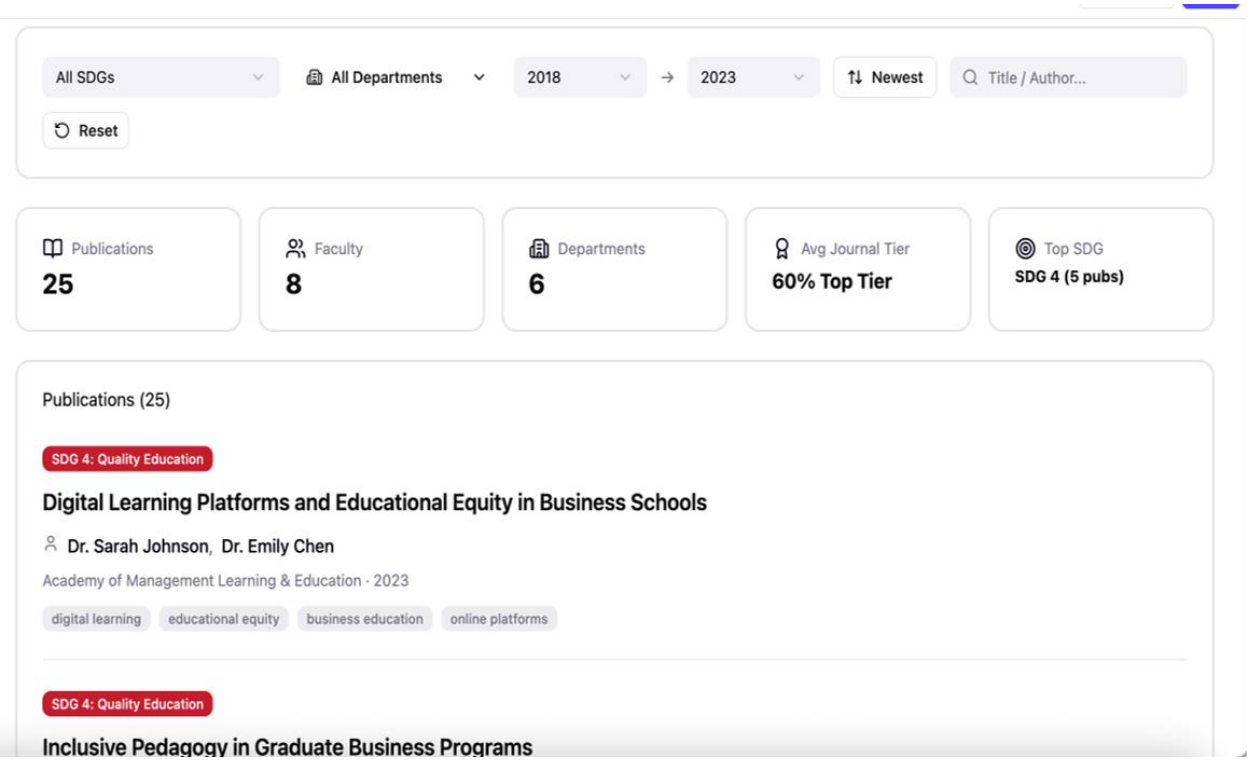


Solution

We introduce a dedicated **SDG Articles Explorer**, allowing students to search and retrieve publications directly.

- Filter by **SDG**, **department**, **author**, or search words in sub-search box
- Summary cards for **publications / faculty / departments / top-tier %**
- Article cards with **SDG tags**, **full metadata**, and **direct access**
- Structured filters enable quick discovery of relevant materials

Students can now find articles they are interested in, find real materials for their homework, projects and proposals without much trouble easily.



SDG Articles Explorer - KPIs + Filter + Directory

Improved Experience

Now, students can seamlessly transition from the broad trends to the specific content of the articles, thus turning this dashboard into a practical tool for finding targeted research.

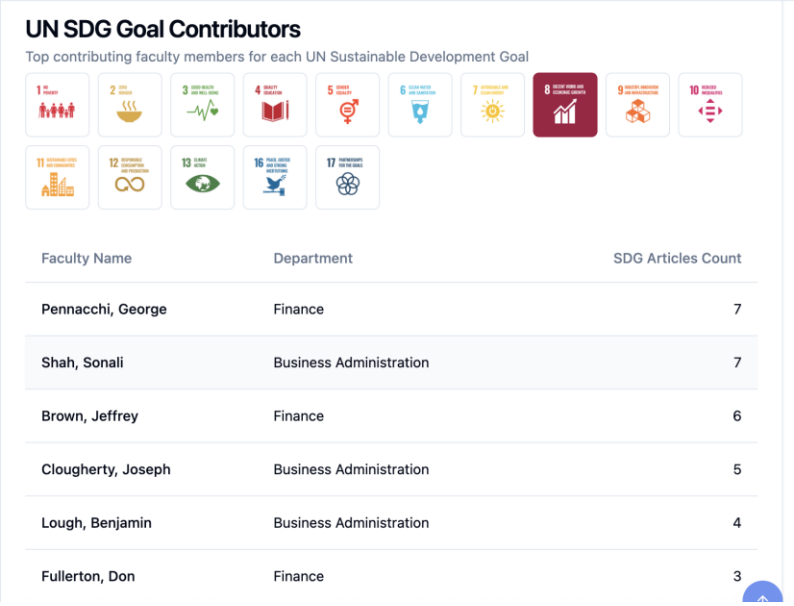
Step 3 – Faculty Identification

Pain Point 1 — Students cannot determine which faculty align with their research interests

Pain Point

After reviewing relevant materials, students start to look for teachers who are dedicated to SDG goals or those topics they want to do further research. The current dashboard only

provides a brief table listing names, departments, and the number of articles published, but lacking citation indicators, measures of research productivity, and an understanding of each professor's academic focus. Students were unable to identify relevant faculties or evaluate who was the most suitable person for seeking guidance, collaboration, or research supervision.

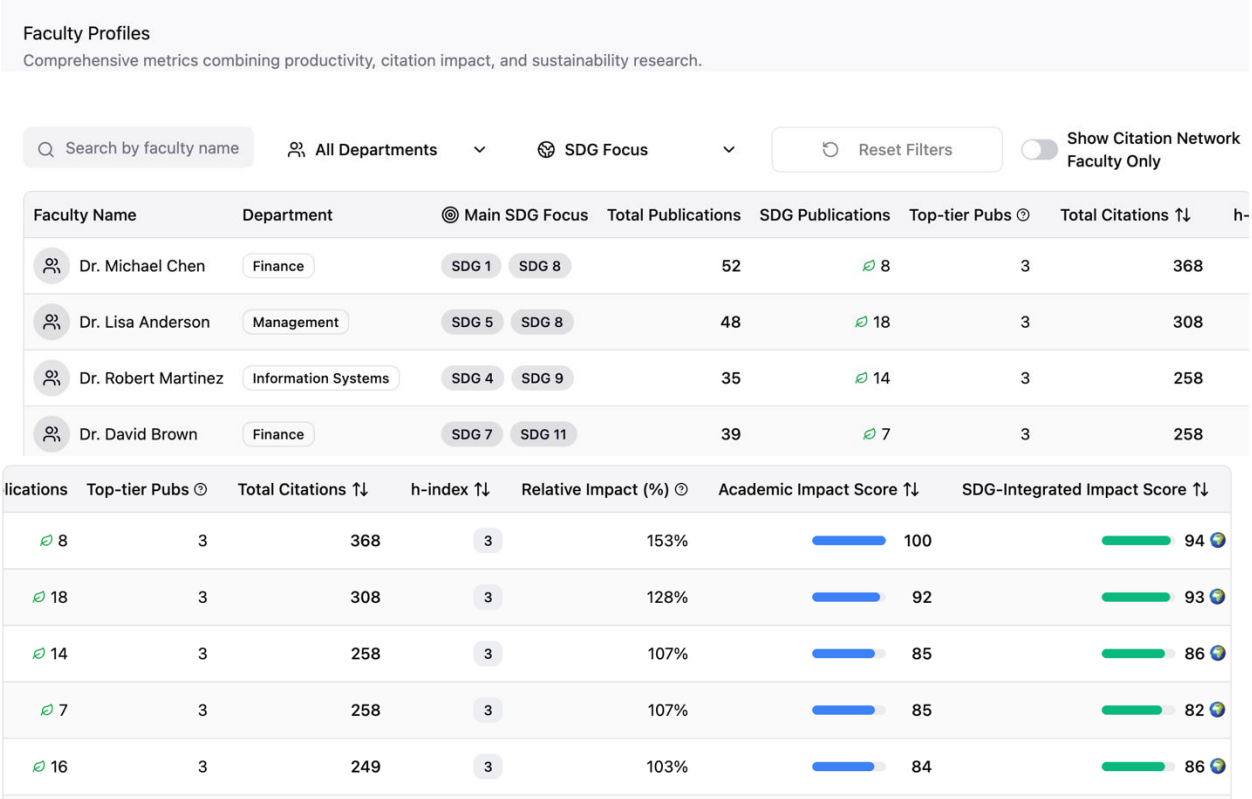


Solution

We design comprehensive Faculty Profiles, aiming to provide a clear structure, comparability and depth. This part includes the following contents:

- publication counts and top-tier journal outputs
- total citation metrics and h-index–style indicators
- SDG focus labels connecting researchers to sustainability themes
- Academic Impact Score and SDG-Integrated Impact Score for broader assessment

Students can filter teachers by department or SDG goals, enabling them to conduct targeted exploration based on their academic goals. This profiles transforms the list of faculties into a navigable research map.



Faculty Impact – Faculty Profiles

Improved Experience

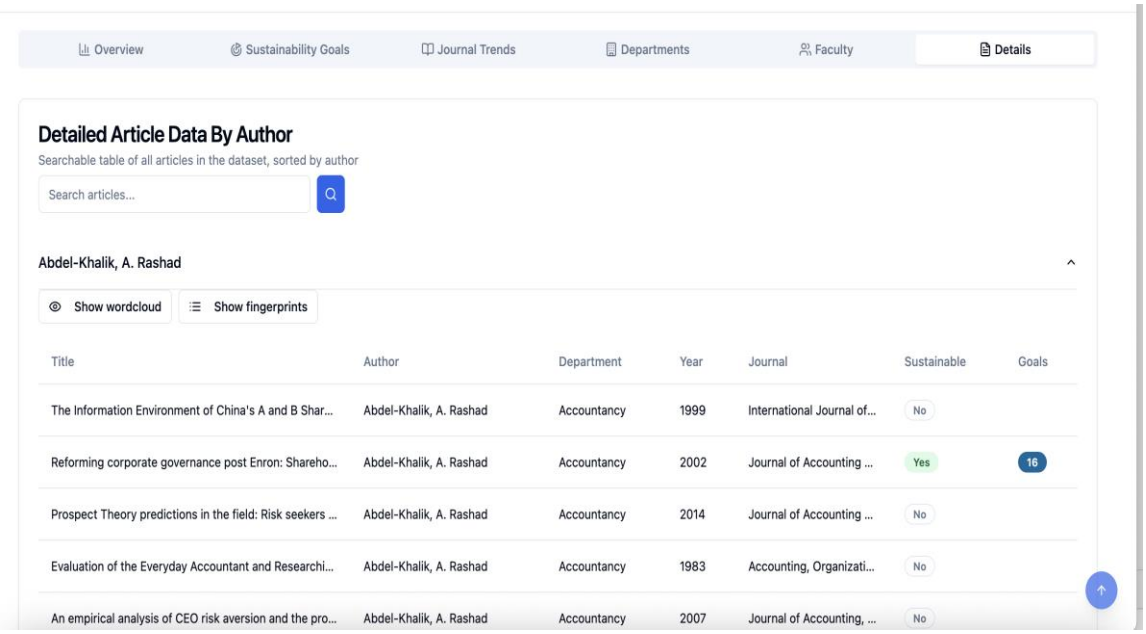
Students can find clear ways to identify faculties whose professional fields align with their own interests. When students choosing potential mentors or partners, they can make informed decisions, reducing uncertainty and enhancing academic compatibility.

Step 4 – Faculty Identifying in Detail

Pain Point 1 — Faculty detail pages do not provide a coherent overview of a professor's research

Pain Point

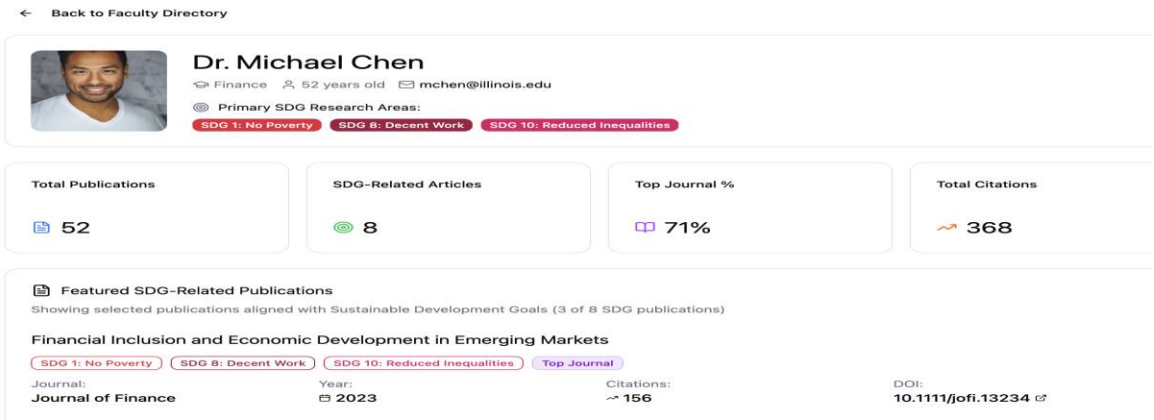
When students choose a specific faculty for a more in-depth review, the original dashboard only displays a long list of articles. Due to the lack of summary indicators such as the total number of published articles, the extent of participation in SDG goals, the number of top journals published, or citation influence, it is difficult to have a clear understanding of a professor's research profile at a glance. Students must manually infer the research topic, influence, and relevance from the original article table, and this process is both time-consuming and prone to errors.

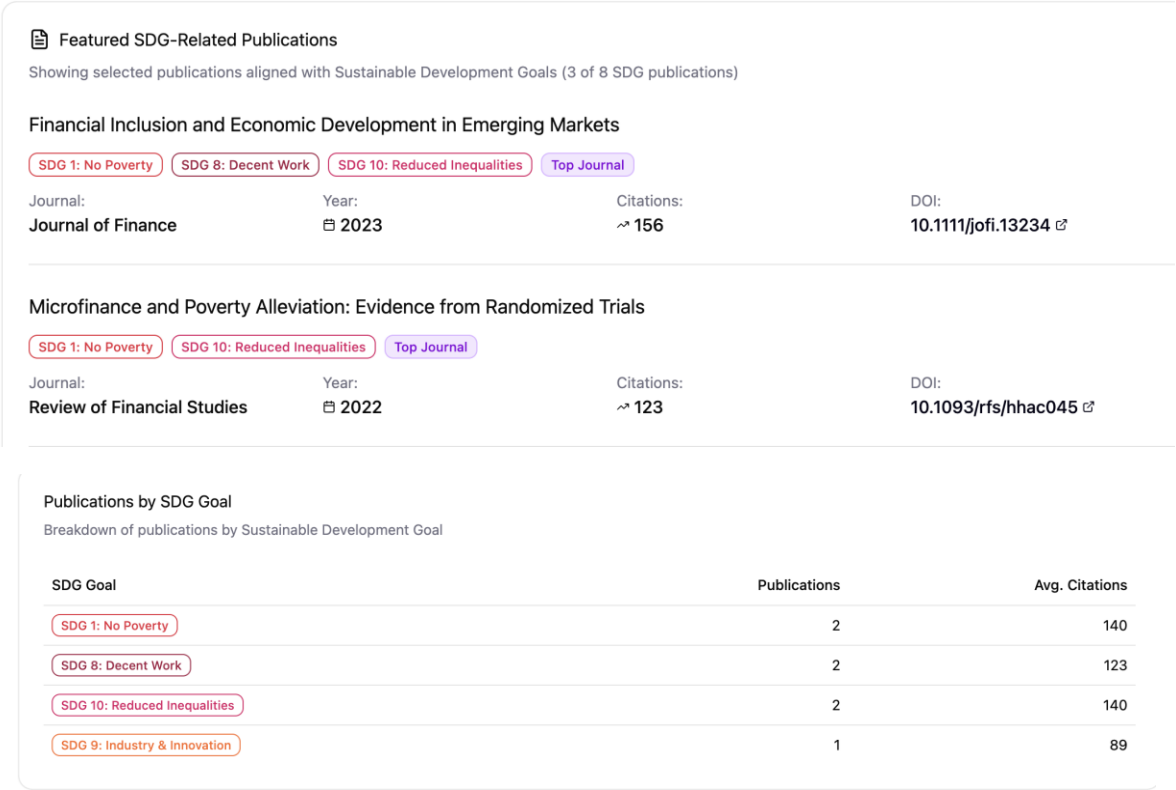


Solution

The newly added **Personal Information page** presents a comprehensive academic overview, with multiple layers of detailed information.

Four summary cards summarize the key indicators: total number of published papers, total numbers of SDG goals, proportion of papers published in top journals, and total citations. Selected papers related to sustainable development goals are highlighted to showcase academic contributions. Additionally, a structured table categorized by goals provides a comprehensive view of the output and average citations in each sustainable development area.





Faculty Impact – Personal Information (mock page, not represents the real faculty)

Improved Experience

This part enables students to understand the research focus, influence, and research trajectory of a college member without having to browse through large unfiltered tables.

This part can effectively assess the compatibility of the research and enhance the ability of students to have meaningful communication with potential mentors.

5.3 Donor Journey

Step1 - Identify High-Impact SDG Goals

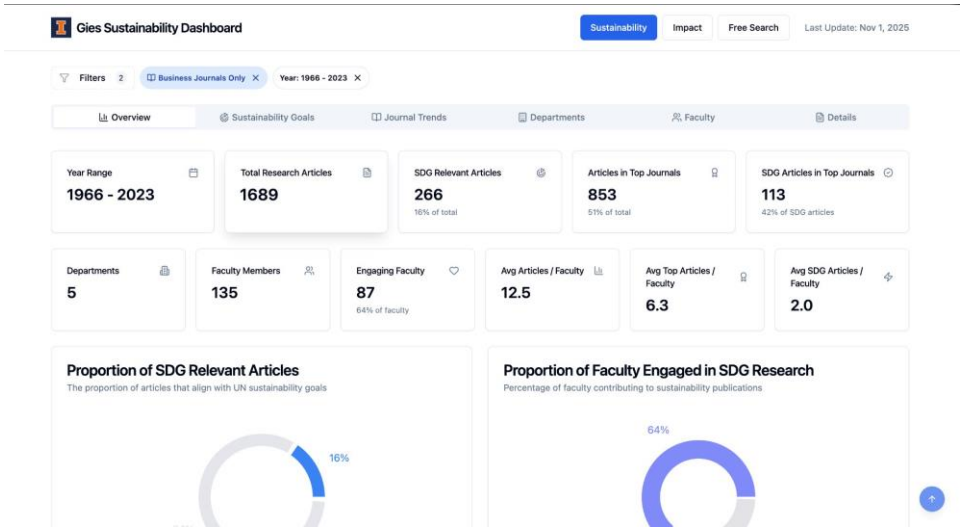
Pain Point 1 — No dedicated donor-focused section

Pain Point

In the current dashboard, donors do not have a designed page for their needs. All the information was embedded in other sections. As a result, donors are unable to quickly

know the impact of their donations, or which areas of sustainable development had the most effective funding.

Due to the lack of a perspective for donors, it was difficult to measure the donation outcomes or evaluate the effectiveness of the funding provided by each university.



Solution

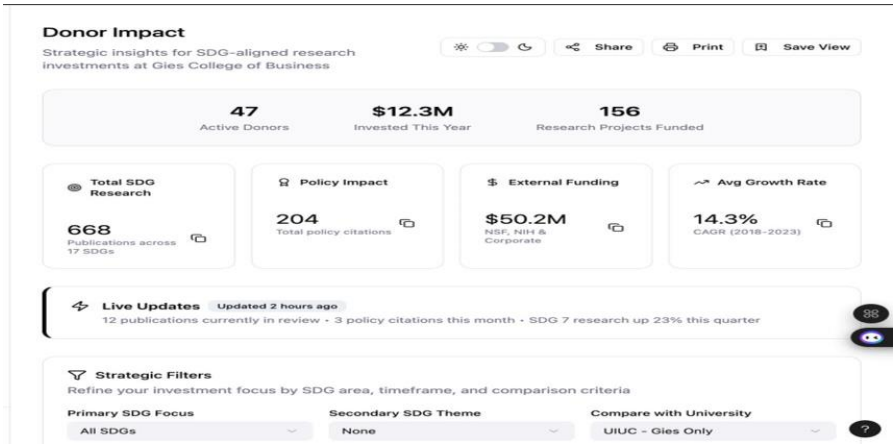
Besides the Home and SDG Research Landscape, we introduce **Donor Impact** section that firstly displays key KPIs donors care about most: total SDG research volume, policy influence, external funding, growth rates, and overall investment outcomes.

The new layout provides:

- A clear, consolidated section of high-impact SDG goals
- Benchmarking capability to compare against peer universities
- Tracking tools to monitor research output, funding flow, and impact growth over time

After donors see the Home and SDG Research Landscape, they can go to Donor Impact. With these insights specific to donors, they can quickly determine which SDG goals will yield the greatest returns, and which areas of further investment will produce the most measurable effects.

When donors visit the Donation Impact, it will present various indicators in a clear and understandable visual format. These indicators can help specific donors make impactful decisions based on data.



Donor Impact – KPIs

Step 2 – Evaluate Potential Impact

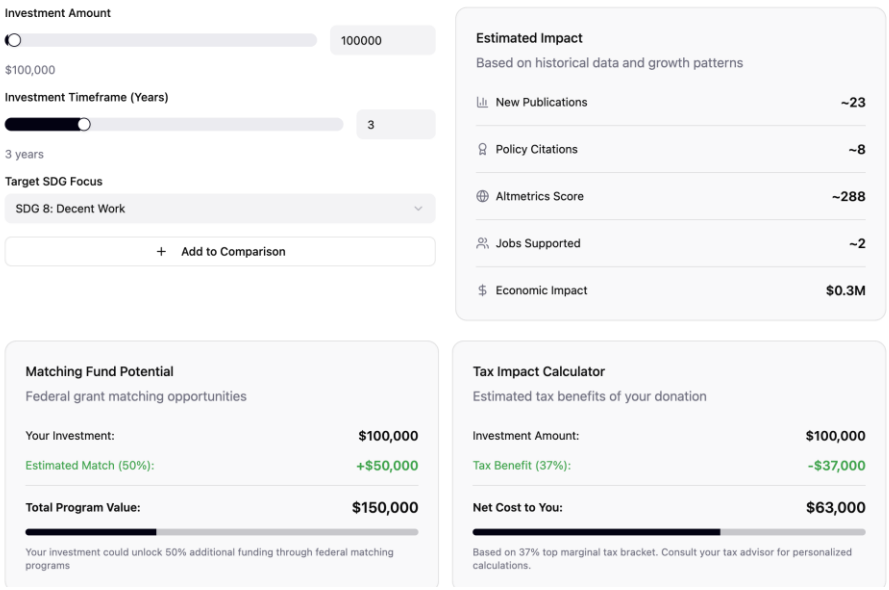
Pain Point 1 — Donors cannot see the concrete return of their funding decisions

Pain Point

In the current dashboard, donors were unable to understand how different amounts of funds were transformed into research outcomes. There were no predictive data, expected returns, visualizations of how the scale of investment would affect social impact.

Solution

The newly designed **Impact Simulator** provides donors with a customizable and data-based predictive tool. Donors can adjust the investment amount, time range, and SDG goals, then can immediately see the expected results, including expected new publications, policy citations, Altmetrics scores, supported jobs, and projected economic impact. This part can also display federal matching opportunities and tax benefit calculations, providing donors with a comprehensive picture of the financial and impact situation. This transforms the abstract donation into a quantifiable and evidence-supported scenario.



Donor Impact – Impact Simulator

Improved Experience

Donors gain a transparent, outcome-oriented view of how their contributions translate into measurable academic and social value. This simulator transforms behaviors into decisions based on expected impacts, financial incentives, and comparative analysis, enabling donors to select investment strategies that yield the maximum returns and align with their charitable goals.

Step 3 – Review Proven Success Stories

Pain Point 1 — lacks persuasive, real-world evidence to justify donor investment

Pain Point

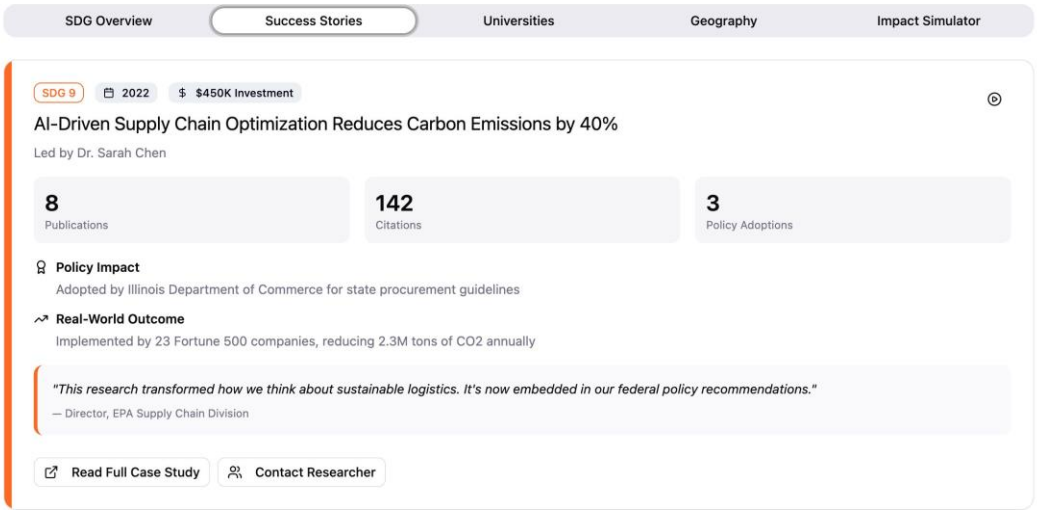
Donors need proof to show their contributions can be generated into meaningful social outcomes. The original dashboard presents publication counts and citations but without **clear narratives**, **policy adoption evidence**, or **real-world impact metrics**. Without concrete stories, donors struggle to evaluate how previous funding translated into social benefit, which may reduce their confidence in making new contributions.

Solution

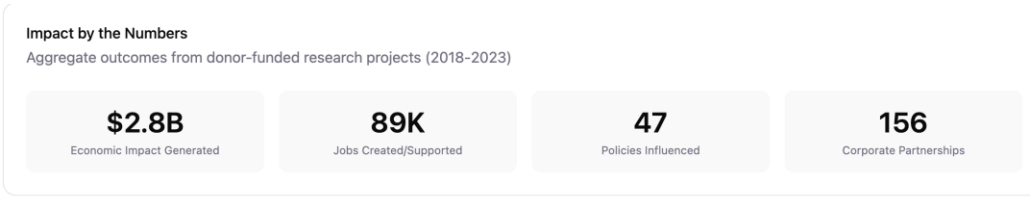
The redesigned **Success Stories** provides donors with rich, verifiable evidence of research impact. Each **Case Study** and **Impact by the Numbers** part highlights:

- **Key outcomes** such as publications, citations, and policy adoptions
- **Real-world implementations** (e.g., Fortune 500 deployment, CO_2 reduction, workforce impact)
- **Economic and social-impact metrics** (e.g., jobs supported, economic value created)
- **Human-centered narratives** featuring quotes from government agencies, industry leaders, or policy partners

This part transforms the original research findings into compelling stories of change, clearly shows how the projects funded have impacted policies and communities.



Donor Impact – Success Stories – Case Study



Donor Impact – Success Stories – Impact by the Numbers

Improved Experience

Donors can obtain reliable evidence of the value generated by their contributions. By observing specific policy changes, measurable outcomes, and real-world success stories, donors will feel more confident, inspired, and more willing to engage in future research projects.

Step 4 – Explore State-Level SDG Performance

Pain Point 1 — Donors cannot see where research impact is geographically concentrated

Pain Point

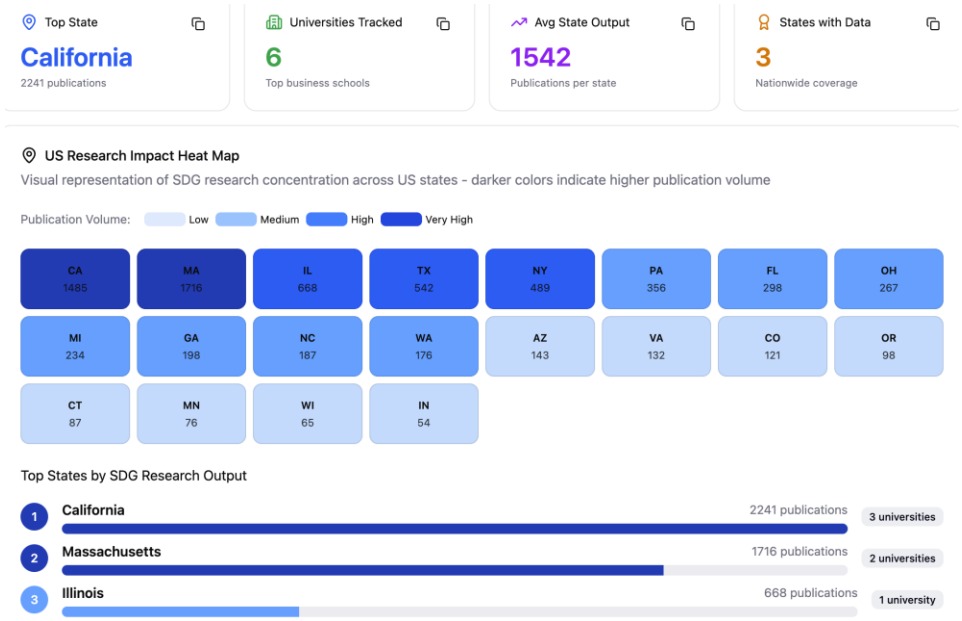
The current dashboard provides no geographical information. Donors are unable to determine which states have the most significant results in SDG goals, which regions dominate specific SDG goals, or the comparison of Gies with other similar institutions in the US. Donors are unable to verify whether their investments were in line with national research trends or the actual needs of the regions.

Solution

The redesigned **Geography** part introduces a multi-layer geographic comparison module.

- A US Research Impact Heat Map highlights publication intensity by state, enabling donors to quickly identify national research clusters.
- A second heat map breaks down SDG performance by topic across the top research states, revealing region-specific strengths (e.g., SDG 9 dominance in California and SDG 8 concentration in Massachusetts).
- University location cards contextualize institutional performance within each state.
- Regional policy-impact summaries show real-world influence at the state and federal levels.

Together, these tools give donors a clear, spatially grounded understanding of national SDG research landscapes.



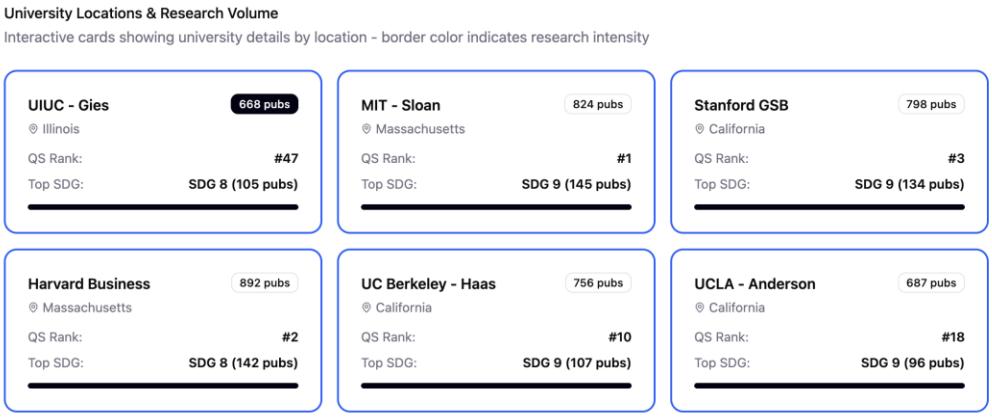
Donor Impact – Geography – US Research Impact Heat Map

SDG Research Heat Map by State

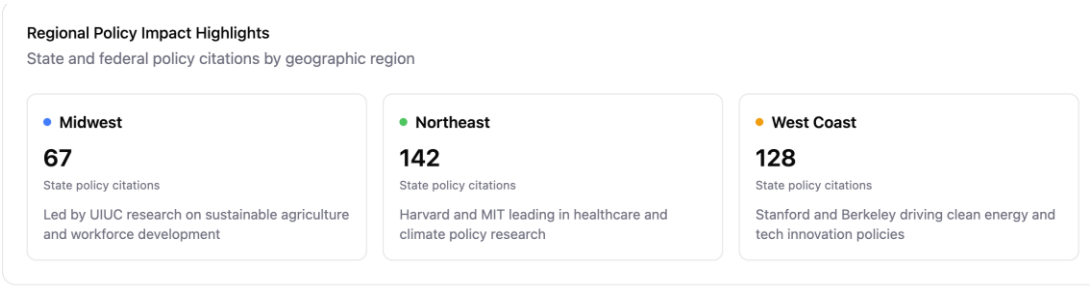
Distribution of top 5 SDG areas across leading research states - hover over cells for details

State	SDG 8	SDG 9	SDG 12	SDG 13	SDG 4
Massachusetts	260	273	192	130	133
California	307	341	271	256	166
Illinois	105	91	73	48	47
Texas	87	76	62	41	38
New York	78	82	54	67	44

Donor Impact – Geography – SDG Research Heat Map by State



Donor Impact – Geography – University Locations & Research Volume



Donor Impact – Geography – Regional Policy Impact Highlights

Improved Experience

Donors can gain a clearer understanding of which regions have outstanding research achievements and the positioning of Gies at the national level. They can verify whether their funding goals align with the strengths of the regions, the policy influence, and the broader national research model, to support more strategic and evidence-based donation decisions.

Step 5 – Explore University-Level SDG Performance

Pain Point 1 — Donors cannot evaluate Gies' SDG competitiveness in context

Pain Point

Donors usually make funding decisions based on the overall strength of each institution.

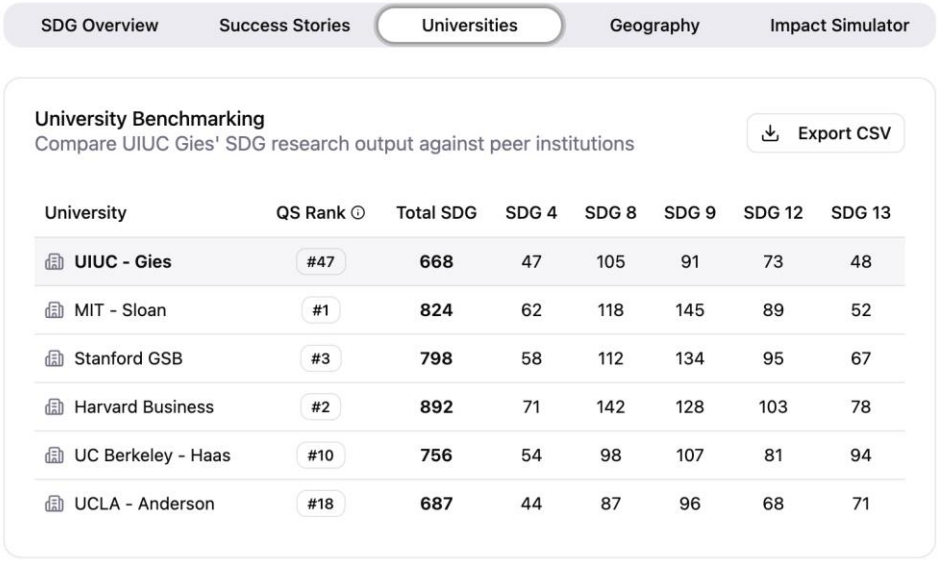
However, the current dashboard does not provide a method to compare Gies with other business schools. Since donors are unable to understand Gies's performance in key SDG goals compared to other schools, they lack evidence to assess its competitiveness, justify the investment, or understand Gies global standing. This made funding decisions uncertain.

Solution

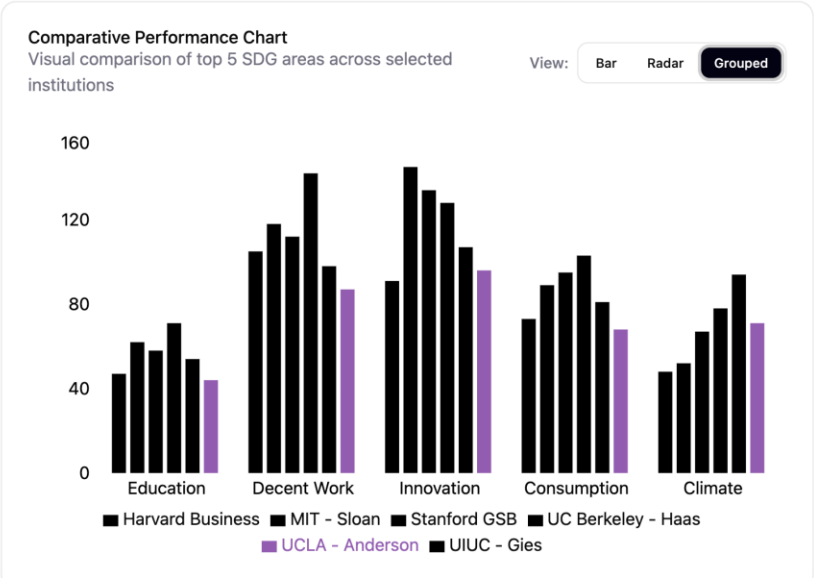
The newly added **Universities** part in Donor Impact introduces a dedicated institutional benchmarking tool.
Donors can now:

- Compare Gies SDG output with peer schools.
- Review both overall SDG volume and goal-level strength.
- Use visual comparative charts (bar / radar / grouped) to show performance differences across SDG goals.
- Validate whether Gies demonstrates distinctive strengths, emerging growth areas, or gaps relative to the competition.

This comparison provides a clear, objective basis for understanding Gies market position in sustainability research.



Donor Impact – Universities – University Benchmarking



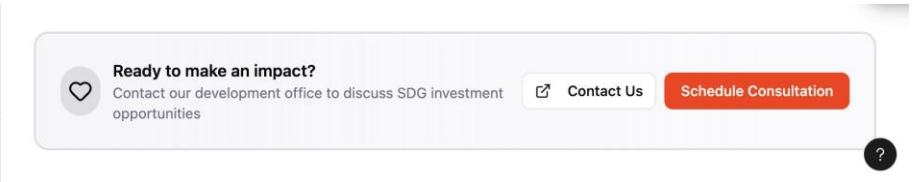
Donor Impact – Universities – Comparative Performance Chart

Improved Experience

Donors can obtain reliable opinions regarding the research competitiveness of Gies. They can assess whether the college is a worthy investment target confidently. They can also support this assessment through transparent comparisons with similar institutions.

Added Feature: Donor Consultation Module

To address the issue of the lack of donor-oriented interactive functions in the original dashboard, we add a simple but crucial module: Ready to Make a Difference? This part provides direct communication channels with Gies through "Contact Us" and "Schedule Consultation" options, offering donors a clear path of action.



Donor Impact – Consultation

6 Users' Journey

6.1 User Flows – Leaders

Step	What the User Wants to Do	User Action	Section	Why It Matters
1 Assess Gies's research situation	See general and SDG research situation and trend in Gies	Opens Home → Reviews key KPIs → SDG Research Landscape	Home & SDG Research Landscape	Helps leaders form a clear strategic understanding of Gies's sustainability identity and where strengths currently are.
2 Strategic Benchmarking	compare the academic outputs of Gies with other universities	Opens Compare Universities → Selects peers → Reviews structural & momentum differences	Compare Universities	Enables leaders to contextualize Gies's position, justify strategic priorities, and identify competitive gaps or advantages.
3 Performance Identification	identify which faculty and department is worthy to being funded for the SDG	Opens Faculty & Department Impact pages → See SDG performance	Faculty Impact & Research Impact	Connects sustainable performance to organizational units, helping leaders direct resources toward impactful scholars and departments.
4 Narrative Validation	extract real, occurred success cases as the basis for external narratives to attract funding and enhance the reputation	Opens Donor Impact → Reviews Success Stories metrics → Reads case documents	Donor Impact-Success Stories	By demonstrating actual influence and policy outcomes, helps to build reputation and facilitate fundraising.
5 Simulated Outcome Assessment	assess the potential return on investment to confirm the amount of	Uses Impact Simulator → Adjusts budget & timeframe → Reviews	Donor Impact-Impact Simulator	Allows quantified justification for funding decisions by predicting how investment drives sustainable outcomes.

	resources they should distribute on each project	estimated impact outputs		
6 Resource Prioritization & Final Verification	make the final decision on the resource prioritization	Reviews insights from Steps 1–5 → Confirms highest-impact SDG focus areas → Makes allocation decisions	All the dashboard	Ensures the final decision is strategically aligned, evidence-backed, and optimized for long-term sustainable impact.

Leaders begin by orienting themselves with high-level KPIs to understand Gies's overall sustainability strengths and SDG identity.

Next, they compare with other institutions to assess Gies's competitiveness, unique features, and areas that need improvement. In this context, leaders will determine investment opportunities by examining the contributions at the department and faculty levels that align with the strategic SDG goals.

To support fundraising and external partnerships, leaders validate these insights through concrete success stories that demonstrate real-world impact. They use the Impact Simulator to estimate potential returns under different funding scenarios, ensuring decisions are evidence-based and financially justified.

Finally, leaders will integrate all the insights, including capability assessment, benchmark comparison, faculty and department assessment, narrative verification, and expected impact assessment, to determine the priority order of resource allocation and confirm the strategic resource allocation plan.

6.2 User Flows – Students

Step	What the User Wants to do	User Action	Section	Why it matters

1 Overview + Understanding SDG Strengths	see the overview and SDG research situation in Gies	Opens the Home → Reads KPIs → SDG Research Landscape	Home & SDG Research Landscape	Gives students a structured understanding of Gies's research identity and primary SDG focus areas.
2 Article Exploration	see specific articles based on some SDG goals they are interested in	Uses filters to Searches articles	SDG Articles Explorer	Provides direct access to relevant publications, enabling students to support assignments, develop research ideas, and explore SDG topics deeply.
3 Faculty Identification	After students browse interested articles, they may have target professors, or want to find more professors based on their interested goals.	Opens Faculty Impact → Faculty Profiles → Filters by department or SDG → Reviews faculty metrics	Faculty Impact	Connects SDG outputs to faculty members, helping students discover relevant researchers and potential mentors.
4 Faculty Identifying in detail	After students find their target professors, they want to see details about professors they interest.	Click one faculty from the profiles → open the Personal Information page → Reviews overview cards → Reads selected SDG-related publications	Faculty – Personal Information	Provides a complete, credible profile of each faculty, allowing students to evaluate alignment, and choose mentors confidently.

Firstly students gain a preliminary understanding by seeing KPI and SDG to gain the research field of Gies and determine the aspects they might be interested in.

Then they start to conduct targeted exploration, browsing the publications based on the themes or issues of SDG goals, in order to quickly collect the relevant materials needed for research or projects.

Once the topic becomes clearer, students will start to look for relevant professors and identify their research topics.

Finally, students will carefully review the detailed information of faculties, such as their research direction, publications in SDG, and prepare for targeted communication or choosing a professor accordingly.

6.3 User Flows – Donors

Step	What the user wants to do	User Action	Section	Why it matters
1 Identify High-Impact SDG Goals	Gain a high-level understanding of Gies's sustainability identity, major SDG strengths, and overall research positioning	Home + KPIs → Navigates to SDG Research Landscape → Donor Impact-SDG Overview	Home & SDG Research Landscape & Donor Impact	Provide donors with necessary background information so that they can understand the insights.
2 Evaluate Potential Impact	Understand the projected impact and ROI of different donation amounts and funding durations	Donor Impact → Impact Simulator	Donor Impact – Impact Simulator	Allows donors to make evidence-based decisions by seeing how contributions translate into measurable research, policy, and economic outcomes.
3 Review Proven Success Stories	Review real-world success stories demonstrating how research has translated into policy influence and tangible societal outcomes	Donor Impact → Success Stories	Donor Impact – Success Stories	Help donors build trust and confidence by showing proven impact, making investment decisions more compelling.
4 Explore State-Level SDG Performance	Explore how SDG research output varies across the US states and understand Gies's geographic influence	Donor Impact → Geography	Donor Impact - Geography	Shows donors where research impact is concentrated and which regions may benefit most from future investment.
5 Explore University-	Evaluate how Gies performs relative to peer institutions in	Donor Impact Section →	Donor Impact - Universities	Helps donors validate Gies's competitiveness

Level SDG Performance	key SDG research metrics	Compare Universities		and decide whether the school represents a strong and credible investment opportunity.

Donors firstly learn about the Gies's sustainability positioning, research strengths, and overall impact.

Then they begin to conduct outcome evaluations, using the Impact Simulator to observe how different donation amounts translate into measurable results, including economic impact, the number of jobs supported, and policy influence.

Next, donors will verify their credibility through actual cases. They will review those success stories that demonstrate how past investments have led to policy implementation, enterprise implementation, and social benefits.

As confidence grows, donors deepen their understanding of the situation through regional insights. They see the concentration of SDG research in various states in the US and identify regions with the strongest potential for influence. They also see Gies' influence across the country.

Finally, the donor benchmarks Gies against peer institutions, using institutional comparison to confirm competitiveness and ensure their investment supports a leading, high-impact research project.

7 Conclusion

All the redesigns transform the Gies Sustainability Dashboard from a static information display into a user-driven and decision-support system. By improving or adding sections based on users' needs, the dashboard now delivers clarity, efficiency, and strategic value.

Each improvement satisfies user tasks and reduces the obstacles in their workflow. The redesigned structure is both broad and deep, providing both overviews and details for actions.

In summary, our improvements provide a more intuitive and evidence-based way to better demonstrate Gies's influence in sustainable development.